

# **Companion to Bird & de Moor, Algebra of Programming**

## §10. Greedy Algorithms

### §10.1. Theory

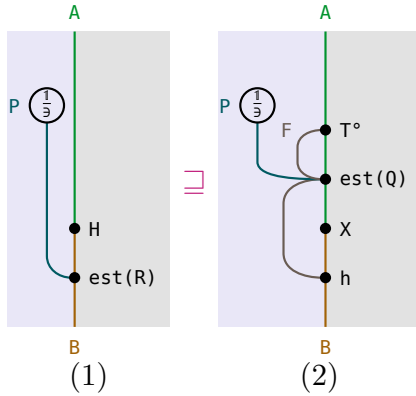
(10.1a)

$h, T, R, H, M$  as in  $M \triangleq \frac{H}{\exists} \text{est}(R)$  ; additionally  $Q$  a **connected** preorder on the sets  $\frac{T^\circ}{\exists}$  returns, so that  $\frac{T^\circ}{\exists} \text{est}(Q)$  is entire.

$\text{Map}(h) \wedge h \text{ Pres } R \wedge R \in R \wedge \text{ThinCondition}(T, h, R, Q) \Rightarrow (\mu X : \frac{T^\circ}{\exists} \text{est}(Q) F(X) h) \sqsubseteq \frac{H}{\exists} \text{est}(R)$  Theorem 10.1

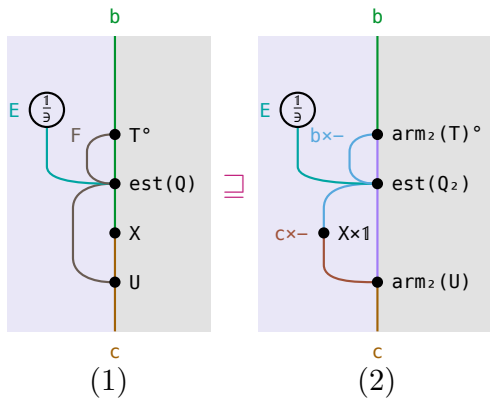
the same optimum reached by keeping ONE decomposition at each step, so that no set is ever carried and the recursion runs on values alone

$H \triangleq (T)^\circ (h) \triangleq H, Q$



|   |                                                            |  |
|---|------------------------------------------------------------|--|
| 1 | $\frac{H}{\exists} \text{est}(R)$                          |  |
| 2 | $\sqsubseteq \frac{T^\circ}{\exists} \text{est}(Q) F(X) h$ |  |

the body at  $T$  a coproduct of two arms

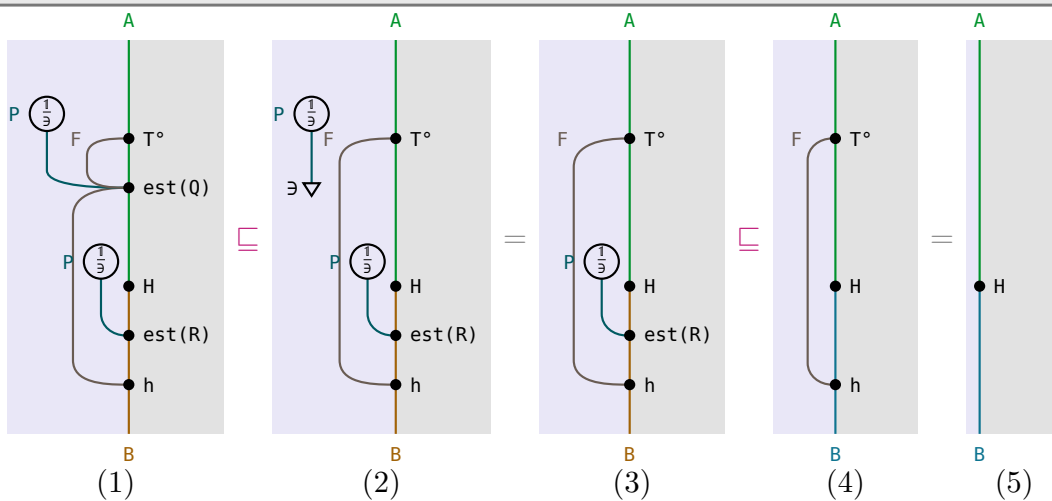


|   |                                                                                                  |                                                           |
|---|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| 1 | $\frac{T^\circ}{\exists} \text{est}(Q) F(X) U$                                                   | Proposition 10.1, $\forall_2 \forall_1^\circ = \emptyset$ |
| 2 | $\sqsubseteq \frac{\text{arm}_2(T)^\circ}{\exists} \text{est}(Q_2) (X \times 1) \text{arm}_2(U)$ |                                                           |

$\frac{T^\circ}{\exists} \text{est}(Q) F(\frac{H}{\exists} \text{est}(R)) h \sqsubseteq H$

(10.1b)

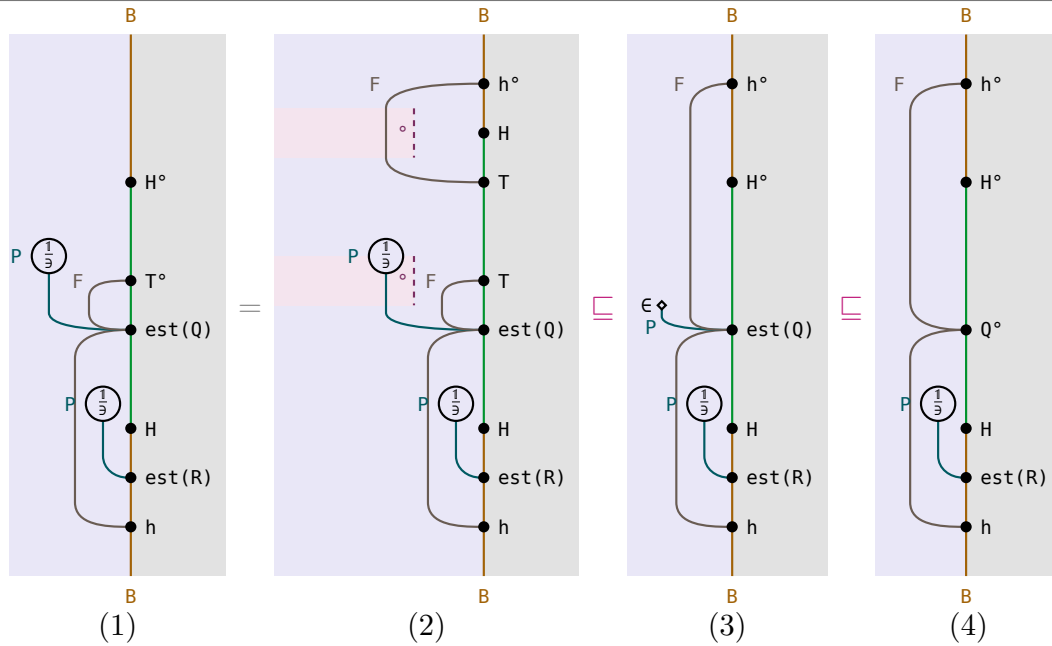
taking the input apart in one  $Q$ -extreme way, solving the parts by  $M$  and assembling by  $h$  returns only what  $H$  returns



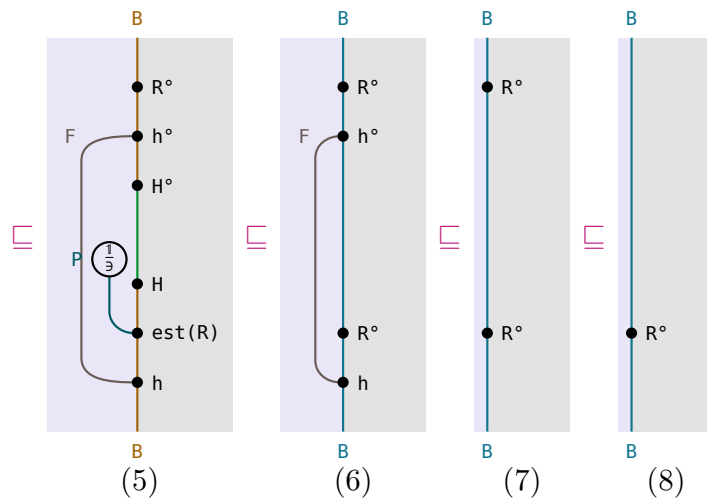
|   |                                                                                    |                                               |
|---|------------------------------------------------------------------------------------|-----------------------------------------------|
| 1 | $\frac{T^\circ}{\exists} \text{est}(Q) F(\frac{H}{\exists} \text{est}(R)) h$       | $\text{est}(Q) \sqsubseteq \exists$           |
| 2 | $\sqsubseteq \frac{T^\circ}{\exists} \exists F(\frac{H}{\exists} \text{est}(R)) h$ | $\frac{R}{\exists} \exists = R$               |
| 3 | $= T^\circ F(\frac{H}{\exists} \text{est}(R)) h$                                   | $\frac{\text{est}(R)}{\exists} \sqsubseteq S$ |
| 4 | $\sqsubseteq T^\circ F(H) h$                                                       | $T^\circ F(H) h = H$                          |
| 5 | $= H$                                                                              |                                               |

$$\text{Map}(h) \wedge h \text{ Pres } R \wedge R R \subseteq R \wedge \text{ThinCondition}(T, h, R, Q) \Rightarrow H^\circ \stackrel{T^\circ}{\subseteq} \text{est}(Q) F\left(\frac{H}{\exists} \text{est}(R)\right) h \subseteq R^\circ \quad (10.1c)$$

$H^\circ$  followed by the greedy body at  $M$  is  $\exists R^\circ$ : an answer of the body is never worse than an answer of  $H$  to the same input



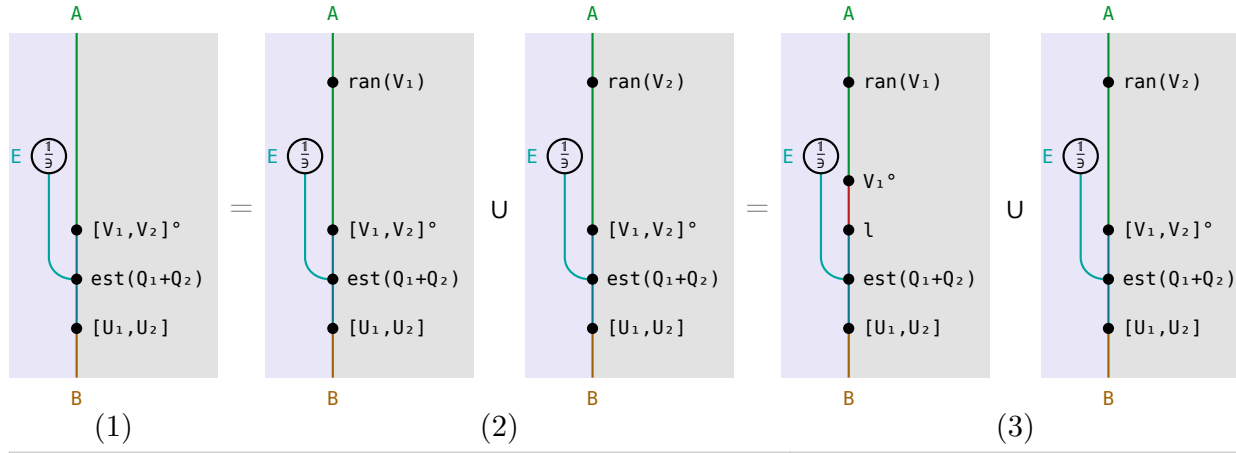
|   |                                                                                                                      |                                                                                                                     |
|---|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 1 | $H^\circ \stackrel{T^\circ}{\subseteq} \text{est}(Q) F\left(\frac{H}{\exists} \text{est}(R)\right) h$                | $H^\circ = h^\circ F(H^\circ) T$                                                                                    |
| 2 | $= h^\circ F(H^\circ) T \stackrel{T^\circ}{\subseteq} \text{est}(Q) F\left(\frac{H}{\exists} \text{est}(R)\right) h$ | $T \stackrel{T^\circ}{\subseteq} \epsilon$                                                                          |
| 3 | $\sqsubseteq h^\circ F(H^\circ) \epsilon \text{est}(Q) F\left(\frac{H}{\exists} \text{est}(R)\right) h$              | $\epsilon \text{est}(R) \subseteq R^\circ$                                                                          |
| 4 | $\sqsubseteq h^\circ F(H^\circ) Q^\circ F\left(\frac{H}{\exists} \text{est}(R)\right) h$                             | $\text{ThinCondition}(T, h, R, Q)$<br>$\Rightarrow h^\circ F(H^\circ) Q^\circ \subseteq R^\circ h^\circ F(H^\circ)$ |



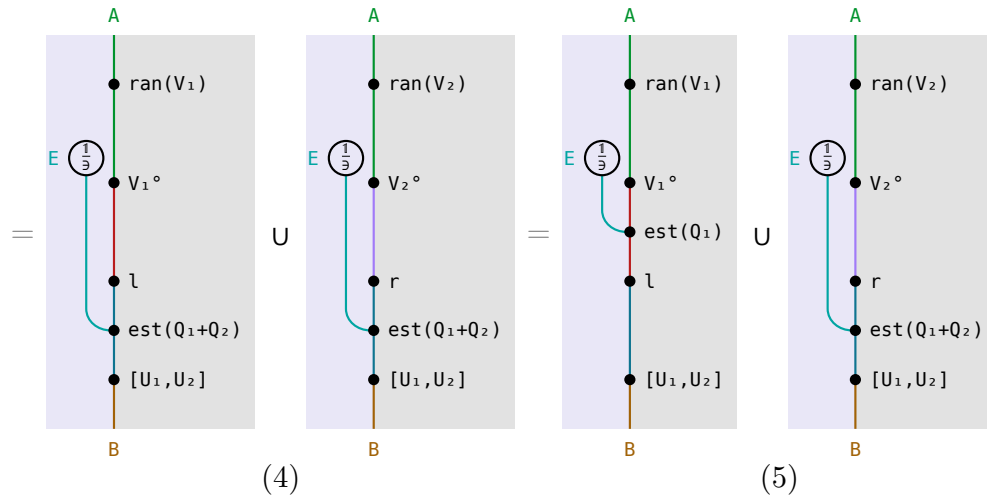
|   |                                                                            |                                                                                                  |
|---|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 5 | $\sqsubseteq R^\circ h^\circ F(H^\circ \frac{H}{\exists} \text{est}(R)) h$ | $S^\circ \stackrel{S}{\subseteq} \text{est}(R) \subseteq R^\circ$                                |
| 6 | $\sqsubseteq R^\circ h^\circ F(R^\circ) h$                                 | $\text{Map}(h) \wedge h \text{ Pres } R$<br>$\Rightarrow h^\circ F(R^\circ) h \subseteq R^\circ$ |
| 7 | $\sqsubseteq R^\circ R^\circ$                                              | $R R \subseteq R$<br>$\Rightarrow R^\circ R^\circ \subseteq R^\circ$                             |
| 8 | $\sqsubseteq R^\circ$                                                      |                                                                                                  |

$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2] = (\text{ran}(V_1) \frac{V_1^\circ}{3} \text{est}(Q_1) U_1) \cup (\text{ran}(V_2) \frac{V_2^\circ}{3} \text{est}(Q_2) U_2) \quad \text{Proposition 10.1}$$

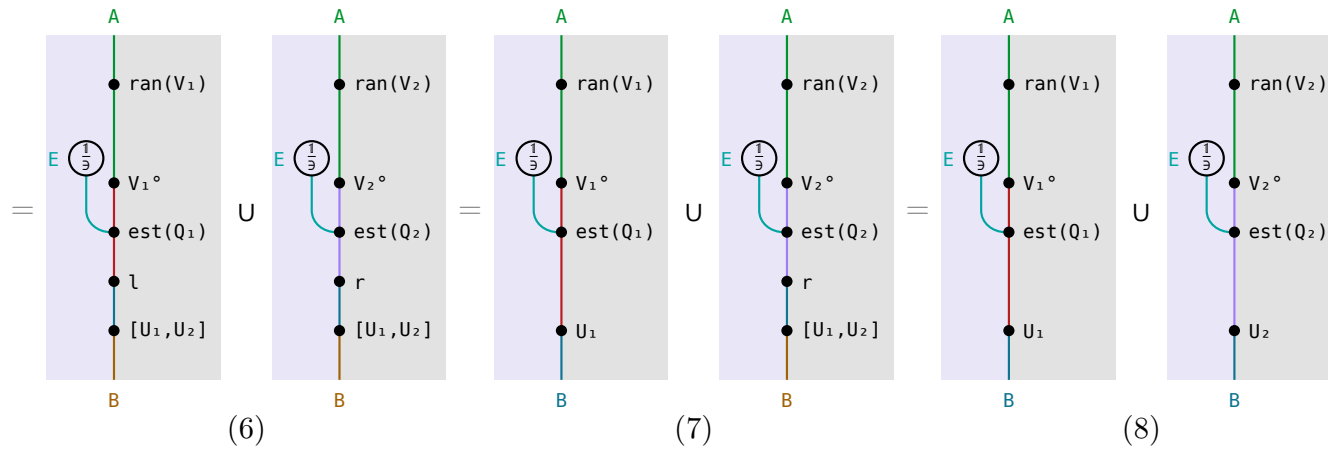
when  $V_1$  and  $V_2$  have disjoint ranges, the greedy step over  $[V_1, V_2]$  runs the  $V_1$  step on inputs  $V_1$  reaches and the  $V_2$  step on inputs  $V_2$  reaches



|   |                                                                                                                                                                      |                                                                                                                                                                                                                                  |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2]$                                                                                                        | $\frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2] = (\text{ran}(V_1) \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2])$ |
| 2 | $= (\text{ran}(V_1) \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2])$ | $V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_1) \frac{[V_1, V_2]^\circ}{3} = \text{ran}(V_1) \frac{V_1^\circ}{3} P(l)$                                                                                                     |
| 3 | $= (\text{ran}(V_1) \frac{V_1^\circ}{3} P(l) \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2])$   | $V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_2) \frac{[V_1, V_2]^\circ}{3} = \text{ran}(V_2) \frac{V_2^\circ}{3} P(r)$                                                                                                     |



|   |                                                                                                                                                                  |                                                  |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| 4 | $= (\text{ran}(V_1) \frac{V_1^\circ}{3} P(l) \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \frac{V_2^\circ}{3} P(r) \text{est}(Q_1 + Q_2) [U_1, U_2])$ | $P(l) \text{est}(Q_1 + Q_2) = \text{est}(Q_1) l$ |
| 5 | $= (\text{ran}(V_1) \frac{V_1^\circ}{3} \text{est}(Q_1) l [U_1, U_2]) \cup (\text{ran}(V_2) \frac{V_2^\circ}{3} P(r) \text{est}(Q_1 + Q_2) [U_1, U_2])$          | $P(r) \text{est}(Q_1 + Q_2) = \text{est}(Q_2) r$ |



|   |                                                                                                                                                |                |
|---|------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 6 | $= (\text{ran}(V_1) \frac{V_1^\circ}{3} \text{est}(Q_1) l [U_1, U_2]) \cup (\text{ran}(V_2) \frac{V_2^\circ}{3} \text{est}(Q_2) r [U_1, U_2])$ | $l [R, S] = R$ |
| 7 | $= (\text{ran}(V_1) \frac{V_1^\circ}{3} \text{est}(Q_1) U_1) \cup (\text{ran}(V_2) \frac{V_2^\circ}{3} \text{est}(Q_2) r [U_1, U_2])$          | $r [R, S] = S$ |
| 8 | $= (\text{ran}(V_1) \frac{V_1^\circ}{3} \text{est}(Q_1) U_1) \cup (\text{ran}(V_2) \frac{V_2^\circ}{3} \text{est}(Q_2) U_2)$                   |                |

## §10.2. The detab-entab problem

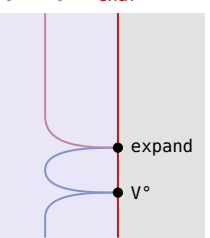
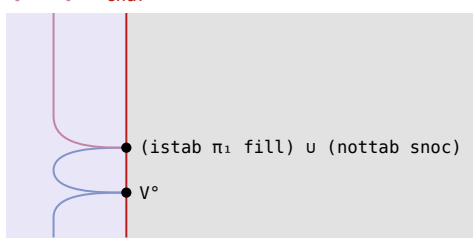
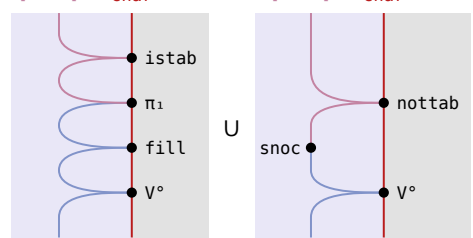
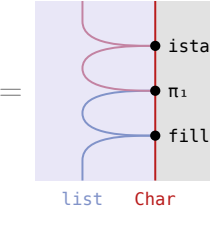
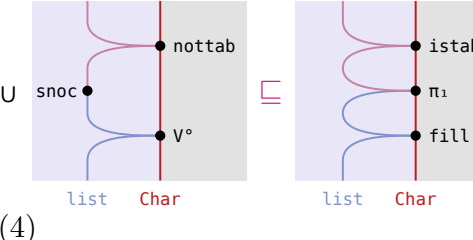
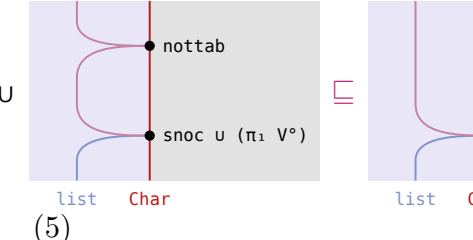
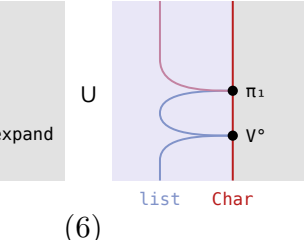
(10.2a)

|    | name          | type                      | definition                                                                                                                                                                                                                                                                                                                                                                                              | meaning                                                                               |
|----|---------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1  | detab         | String → String           | $([nil, expand]) = \text{detab}$ , $\text{detab}(\text{wrap}(a)) = []$ ,<br>$\text{detab}(\text{snoc}(x, a)) = \text{expand}(\text{detab}(x), a)$                                                                                                                                                                                                                                                       | replace every tab by blanks up to the next tab stop; $H = \text{detab}^\circ$         |
| 2  | expand(xs, a) | String                    | $\text{expand}(xs, a) \triangleq$ if $a = \text{tb}$ then fill $xs$ else $\text{snoc } xs \ a$                                                                                                                                                                                                                                                                                                          | a tab fills the line to the next stop; any other character is appended                |
| 3  | fill(xs)      | String                    | $\text{fill}(xs) \triangleq xs ++ \text{blanks } (n - \text{col } xs \% n)$                                                                                                                                                                                                                                                                                                                             | append blanks up to the next multiple of $n$ columns                                  |
| 4  | col()         | String → ℕ                | $\text{col}(\text{wrap}(a)) = 0$ , $\text{col}(\text{snoc}(x, a)) =$ if $a = \text{nl}$ then 0 else $\text{col } x + 1$                                                                                                                                                                                                                                                                                 | the column the string ends at: characters since the last newline                      |
| 5  | R             | [Char] → [Char]           | $xs \ R \ ys \iff \text{length}(xs) \leq \text{length}(ys)$                                                                                                                                                                                                                                                                                                                                             | $x$ is no longer than $y$                                                             |
| 6  | U             | Char → Char               | $a \ U \ b \iff a = \text{tb} \vee a = b$                                                                                                                                                                                                                                                                                                                                                               | $a$ is the tab or $a=b$                                                               |
| 7  | V             | [Char] → [Char]           | $xs \ V \ ys \iff \text{prefix}(xs, ys) \wedge \text{fill}(xs) = \text{fill}(ys)$                                                                                                                                                                                                                                                                                                                       | a prefix that fills to the same string                                                |
| 8  | Q             | F[Char] → F[Char]         | $Q = 1 + (V \times U)$ , $l(\text{val}) \ Q \ l(\text{val}') \iff \text{true}$ , $r(p) \ Q \ r(q) \iff V(\pi_1(p), \pi_1(q)) \wedge U(\pi_2(p), \pi_2(q))$ ,<br>$(\forall \text{val}. \forall \text{val}'. u = l(\text{val}) \implies v = l(\text{val}') \implies \text{false}) \wedge (\forall p. \forall q. u = r(p) \implies v = r(q) \implies \text{false}) \implies (u \ Q \ v \iff \text{false})$ | compare two decompositions by $V$ on the strings and $U$ on the characters            |
| 9  | unfill()      | String → String           | $\text{unfill}(\text{wrap}(a)) = []$ , $\text{unfill}(\text{snoc}(x, a)) =$ if $a = \text{blank} \wedge \text{col } (\text{snoc } x \ a) \% n \neq 0$ then $\text{unfill } x$ else $\text{snoc } x \ a$                                                                                                                                                                                                 | the shortest prefix that fills to the same string                                     |
| 10 | tbc()         | String → ℕ                | $\text{tbc}(\text{wrap}(a)) = 0$ , $\text{tbc}(\text{snoc}(x, a)) =$ if $a = \text{blank} \wedge \text{col } (\text{snoc } x \ a) \% n \neq 0$ then $\text{tbc } x + 1$ else 0                                                                                                                                                                                                                          | the number of trailing blanks                                                         |
| 11 | triple        | [Char] → String × (ℕ × ℕ) | $\text{triple} \triangleq (\text{unfill } \text{entab}, (\text{tbc}, \text{col}))$                                                                                                                                                                                                                                                                                                                      | what <b>entab</b> keeps: the unfilled output, its trailing blank count and its column |

$$\emptyset < n \Rightarrow \text{expand } V^\circ \sqsubseteq \text{expand } u \ (\pi_1 V^\circ)$$

(10.2b)

shortening the output of one **expand** step to a  $V$ -smaller string either keeps the whole step (**expand**) or drops its character and shortens its input string ( $\pi_1 V^\circ$ )

|                                                                                              |                                                                                              |                                                                                               |                                                                                                |
|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
|  <p>(1)</p> |  <p>(2)</p> |  <p>(3)</p> |  <p>(4)</p> |
|  <p>(5)</p> |  <p>(6)</p> |  <p>(7)</p> |  <p>(8)</p> |

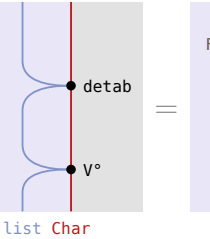
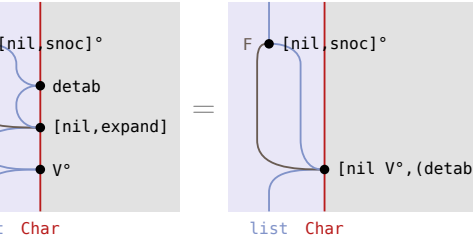
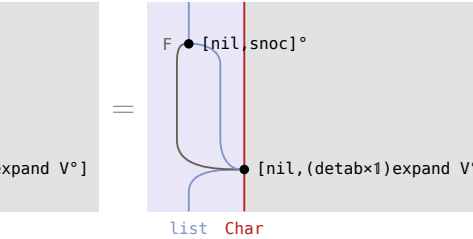
|                                                                                            |                                                                                                    |
|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| $\text{expand } V^\circ$                                                                   | $\text{expand} = (\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } \text{snoc})$             |
| $= ((\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } \text{snoc})) V^\circ$         | $(S \cup T) R = (S R) \cup (T R)$                                                                  |
| $= (\text{istab } \pi_1 \text{ fill } V^\circ) \cup (\text{nottab } \text{snoc } V^\circ)$ | $\emptyset < n \wedge \text{blank} \neq n\ell$<br>$\Rightarrow \text{fill } V^\circ = \text{fill}$ |

|                                                                                                           |                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| $= (\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } \text{snoc } V^\circ)$                         | $\emptyset < n \wedge \text{blank} \neq n\ell$<br>$\Rightarrow \text{snoc } V^\circ \sqsubseteq \text{snoc } u \ (\pi_1 V^\circ)$ |
| $\sqsubseteq (\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } (\text{snoc } u \ (\pi_1 V^\circ)))$ | $(\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } (\text{snoc } u \ X)) \sqsubseteq \text{expand} \cup X$                  |
| $\sqsubseteq \text{expand } u \ (\pi_1 V^\circ)$                                                          |                                                                                                                                   |

$$\emptyset < n \Rightarrow \text{detab } V^\circ \sqsubseteq R^\circ \text{ detab}$$

(10.2c)

any  $V$ -shortening of **detab**'s output is **detab**'s output on an input no longer than the given one

|                                                                                                |                                                                                                |                                                                                                 |                                                                                                  |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
|  <p>(1)</p> |  <p>(2)</p> |  <p>(3)</p> |  <p>(4)</p> |
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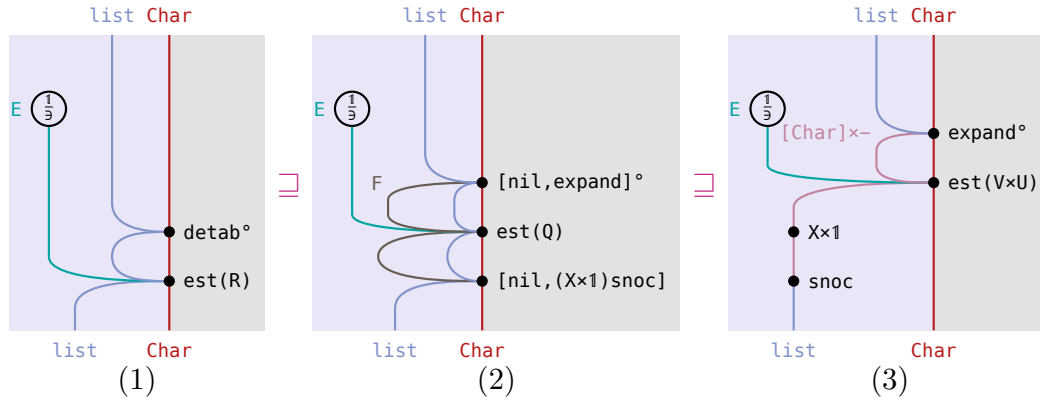
|                                                                                                   |                                                                                                                |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| $\text{detab } V^\circ$                                                                           | $\text{detab} = [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand}]$                   |
| $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand}] V^\circ$           | $[R, S] Z = [R Z, S Z]$                                                                                        |
| $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil } V^\circ, \text{expand } V^\circ]$ | $\text{nil } V^\circ = \text{nil}$                                                                             |
| $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } V^\circ]$          | $(10.2b) \emptyset < n$<br>$\Rightarrow \text{expand } V^\circ \sqsubseteq \text{expand } u \ (\pi_1 V^\circ)$ |

|                                                                                                                |                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\sqsubseteq [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } u \ (\pi_1 V^\circ)]$ | $[\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } u \ X] = \text{detab } u \ (\text{snoc}^\circ (\text{detab} \times 1) X)$ |
| $= \text{detab } u \ (\text{snoc}^\circ (\text{detab} \times 1) \pi_1 V^\circ)$                                | $\text{snoc}^\circ (\text{detab} \times 1) \pi_1 Y = \text{snoc}^\circ \pi_1 \text{ detab } Y$                                                         |
| $= \text{detab } u \ (\text{snoc}^\circ \pi_1 \text{ detab } V^\circ)$                                         | $\emptyset < n$<br>$\Rightarrow \text{detab } u \ (\text{snoc}^\circ \pi_1 \text{ detab } V^\circ) \sqsubseteq R^\circ \text{ detab}$                  |
| $\sqsubseteq R^\circ \text{ detab}$                                                                            |                                                                                                                                                        |

$$0 < n \Rightarrow (\mu X : \frac{[nil, expand]^{\circ}}{3} \text{ est}(Q) F(X) [nil, snoc]) \sqsubseteq \frac{detab^{\circ}}{3} \text{ est}(R) \quad (10.2d)$$

the shortest input **detab** expands to the given output is one pass along that output, holding each blank back and cashing the held blanks in for a tab wherever the column reaches a tab stop

the specification —  $\text{detab}(\text{wrap}(a)) = []$ ,  $\text{detab}(\text{snoc}(x, a)) = \text{expand}(\text{detab}(x), a)$ ;  $\text{detab entab} = 1$  and nothing shorter does



$$1 \quad \frac{detab^{\circ}}{3} \text{ est}(R)$$

at  $Q \triangleq 1 + (V \times U)$ : one character of input is decided at each step.  $F(\tau, R) \alpha \sqsubseteq \alpha R$ .  $\text{detab prefix} \sqsubseteq R^{\circ}$   $\text{detab}$  is FALSE, — at  $n=8$ ,  $\text{detab } [a, b, c, d, e, TB] = [a, b, c, d, e, BL, BL, BL]$ , whose prefix  $[a, b, c, d, e, BL, BL]$  is longer than any input giving it, and  $\text{detab } V^{\circ} \sqsubseteq R^{\circ}$   $\text{detab}$  holds.  $\text{expand } V^{\circ} \sqsubseteq \text{expand } U$  ( $\pi_1 V^{\circ}$ ) — shortening the output either leaves the last step alone or discards it

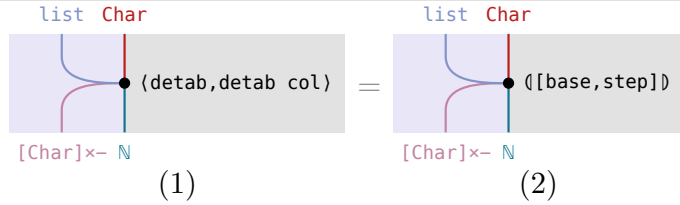
$$2 \quad \sqsubseteq \frac{[nil, expand]^{\circ}}{3} \text{ est}(Q) F(X) [nil, snoc]$$

Proposition 10.1: **nil** and **expand** have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop

$$3 \quad \sqsubseteq \frac{expand^{\circ}}{3} \text{ est}(V \times U) (X \times 1) \text{ snoc}$$

$$\langle \text{detab}, \text{detab col} \rangle = \langle [base, step] \rangle \quad (10.2e)$$

one pass over the input carries the output so far together with its column



$$1 \quad \langle \text{detab}, \text{detab col} \rangle$$

both components are folds over the same input; **step** on  $(\text{detab } x, \text{col}(\text{detab } x))$  is  $(\text{detab}, \text{col detab})$  one character further — not a tabulated row

$$2 \quad = \langle [base, step] \rangle$$

Exercise 10.1

$$\bullet \quad \langle [base, step] \rangle (\text{convert}(xs), r) \Leftrightarrow r = \text{loop}(\text{step}, ([], 0), xs)$$

the fold of  $[base, step]$  over the input converted to a snoc-list is the left loop of **step** from **base**

$$\bullet \quad \pi_1(\text{loop}(\text{step}, ([], c_0), xs)) = \text{loop}'((f, g), (c_0, xs))$$

Exercise 10.1: when **step** only appends  $f(c, a)$  to the output and moves the state by  $g$ , the output of the loop is  $\text{loop}'(f, g)$ , which never carries the output

$$\bullet \quad a \neq \text{tb} \Rightarrow \text{expand}(ys, b) = \text{snoc}(xs, a) \Leftrightarrow \text{fill}(ys) = \text{snoc}(xs, a) \wedge b = \text{tb} \vee (ys, b) = (xs, a)$$

**expand** produces  $x \# [a]$  from a tab after a string that fills to it, or from  $x$  and  $a$

$$\bullet \quad 0 < n \Rightarrow (\exists ys. \text{fill}(ys) = \text{snoc}(xs, a)) \Leftrightarrow a = \text{blank} \wedge \text{col}(\text{snoc } xs \ a) \% n = 0$$

a string is a **fill** exactly when it ends in a blank on a tab stop

$$\bullet \quad \text{entab}(xs) = \text{entab}(\text{unfill } xs) ++ \text{blanks}(\text{tbc } xs) \quad (10.1)$$

(10.1): the output of **entab** is that of **unfill**  $x$  followed by the trailing blanks held back

$$\bullet \quad \text{tbc}[] = 0$$

the empty string has no trailing blanks

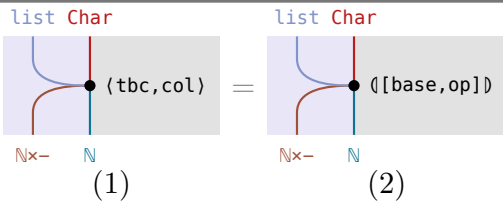
$$\bullet \quad \text{tbc}(\text{snoc}(xs, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc } xs \ a) \% n \neq 0 \text{ then } \text{tbc } xs + 1 \text{ else } 0$$

a blank off a tab stop adds one held blank; anything else releases them all

**(tbc,col) = ([base,op])**

(10.2g)

one pass counts the held blanks and the column together



1 (tbc,col)

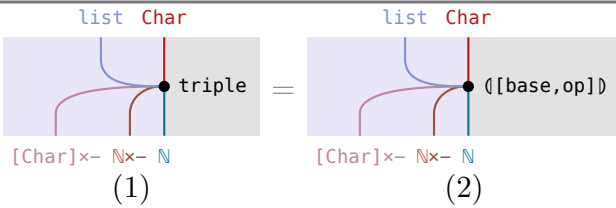
2 = ([base,op])

op on (tbc x,col x) is (tbc,col) one character further, from the equations for tbc — (10.1)  $\text{entab}(xs) = \text{entab}(\text{unfill } xs) ++ \text{blanks } (\text{tbc } xs)$ ,  $\text{tbc}[] = 0$ ,  $\text{tbc}(\text{snoc}(xs,a)) = \text{if } a = \text{blank} \wedge \text{col } (\text{snoc } xs \ a) \% n \neq 0 \text{ then } \text{tbc } xs + 1 \text{ else } 0$

**triple = ([base,op])**

(10.2h)

one pass carries the output up to the held blanks, their count and the column



1 triple

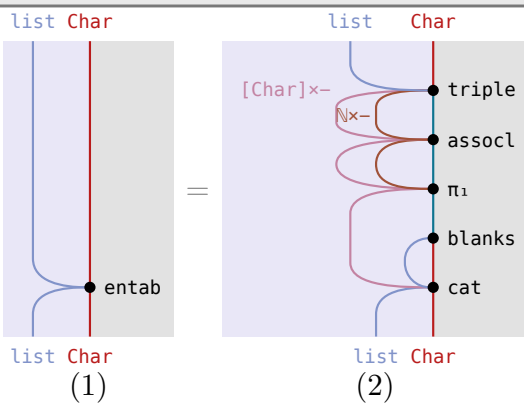
2 = ([base,op])

op on triple x is triple one character further: (10.1) where the blanks are released — (10.1)  $\text{entab}(xs) = \text{entab}(\text{unfill } xs) ++ \text{blanks } (\text{tbc } xs)$ ,  $\text{tbc}[] = 0$ ,  $\text{tbc}(\text{snoc}(xs,a)) = \text{if } a = \text{blank} \wedge \text{col } (\text{snoc } xs \ a) \% n \neq 0 \text{ then } \text{tbc } xs + 1 \text{ else } 0$

**entab = triple assocl  $\pi_1$  (1 × blanks) cat**

(10.2i)

entab is triple with the held blanks appended to its output



1 entab

2 = triple assocl  $\pi_1$  (1 × blanks) cat

(10.1) — (10.1)  $\text{entab}(xs) = \text{entab}(\text{unfill } xs) ++ \text{blanks } (\text{tbc } xs)$ ,  $\text{tbc}[] = 0$ ,  $\text{tbc}(\text{snoc}(xs,a)) = \text{if } a = \text{blank} \wedge \text{col } (\text{snoc } xs \ a) \% n \neq 0 \text{ then } \text{tbc } xs + 1 \text{ else } 0$

## §10.3. The minimum tardiness problem

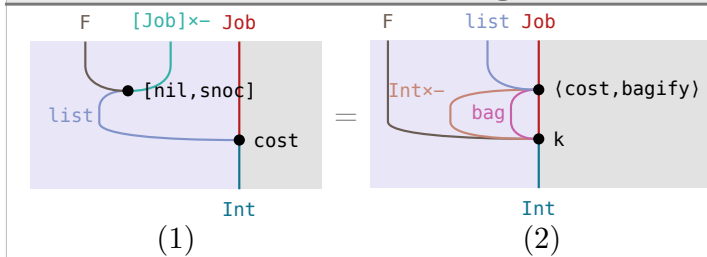
(10.3a)

|    | name                        | type                                         | definition                                                                                                                                                                                                                    | meaning                                                                                                                                                                                                                                                                                                                                                |
|----|-----------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <code>bag(Job)</code>       | $\mathcal{A}$                                | $\text{bag}(\text{Job}) \triangleq \text{Quotient}(\text{perm})$                                                                                                                                                              | a bag of jobs: a list up to reordering                                                                                                                                                                                                                                                                                                                 |
| 2  | <code>snag(p)</code>        | <code>bag(Job)</code>                        | $\text{snag}(p) \triangleq \pi_2 p :: \pi_1 p$                                                                                                                                                                                | put one job into a bag                                                                                                                                                                                                                                                                                                                                 |
| 3  | <code>bagify</code>         | <code>[Job] → bag(Job)</code>                | $\emptyset[\text{nil}, \text{snag}] = \text{bagify}, \text{bagify}(\text{wrap}(u)) = \text{nil}, \text{bagify}(\text{snoc}(xs, a)) = \text{snag}(\text{bagify}(xs), a)$                                                       | forget the order of a schedule; $H = \text{bagify}^\circ$ lists every schedule of a bag                                                                                                                                                                                                                                                                |
| 4  | <code>penalty(xs, j)</code> | <code>Int</code>                             | $\text{penalty}(xs, j) \triangleq (\text{sum}(\text{list}(\text{ct})) \text{ } xs + \text{ct } j - \text{dt } j) \text{wt}(j)$                                                                                                | how late job $j$ finishes after the schedule $xs$ , times its weight                                                                                                                                                                                                                                                                                   |
| 5  | <code>cost</code>           | <code>[Job] → Int</code>                     | $\text{cost}(\text{wrap}(a)) = 0, \text{cost}(\text{snoc}(x, a)) = \text{bmax}(\text{cost}(x), \text{penalty}(x, a))$                                                                                                         | the largest penalty of any job in the schedule                                                                                                                                                                                                                                                                                                         |
| 6  | <code>bmax(a, b)</code>     | <code>Int</code>                             | $\text{bmax}(a, b) \triangleq \text{if } a \leq b \text{ then } b \text{ else } a$                                                                                                                                            | the larger of two numbers                                                                                                                                                                                                                                                                                                                              |
| 7  | $R$                         | <code>[Job] → [Job]</code>                   | $R = \text{cost} \leq \text{cost}^\circ, R(xs, ys) \iff \text{cost}(xs) \leq \text{cost}(ys)$                                                                                                                                 | $xs$ costs no more than $ys$                                                                                                                                                                                                                                                                                                                           |
| 8  | <code>add</code>            | <code>[Job] × Job → [Job]</code>             | $p \text{ add } ys \iff \text{add}(\pi_2(p), ys)$                                                                                                                                                                             | insert $j$ anywhere in $xs$ ;<br>$\text{perm} \triangleq \text{bagify}$<br>$\text{bagify}^\circ = \emptyset[\text{nil}, \text{add}]$                                                                                                                                                                                                                   |
| 9  | $k$                         | <code>F(Int × bag(Job)) → Int</code>         | $k \triangleq [\text{nil}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{ penalty})) \text{ bmax}], k(l(u)) = 0, k(r((c, xs), j)) = \text{bmax}(c, (\text{bagify}^\circ \times 1) \text{ penalty}((xs, j)))$ | the cost of a schedule from the cost of its front and the penalty of its last job                                                                                                                                                                                                                                                                      |
| 10 | $f$                         | <code>F(bag(Job)) → Int</code>               | $f(l(\text{val})) = 0, f(r(p)) = (\text{bagify}^\circ \times 1) \text{ penalty}(p)$                                                                                                                                           | the penalty of the last job, read off the bag before it                                                                                                                                                                                                                                                                                                |
| 11 | $Q$                         | <code>F(bag(Job)) → F(bag(Job))</code>       | $u Q v \iff f(u) \leq f(v)$                                                                                                                                                                                                   | the last job of $u$ gets no larger penalty than that of $v$                                                                                                                                                                                                                                                                                            |
| 12 | $Q'$                        | <code>bag(Job) × Job → bag(Job) × Job</code> | $p Q' q \iff (\text{bagify}^\circ \times 1) \text{ penalty}(p) \leq (\text{bagify}^\circ \times 1) \text{ penalty}(q)$                                                                                                        | $Q$ on the <code>snag</code> summand: compare the penalties of the two last jobs                                                                                                                                                                                                                                                                       |
| 13 | $g$                         | <code>F[Job] → Int</code>                    | $g \triangleq [\text{zero}, \text{penalty}], g(s, c) \iff c = g(s), g(l(u)) = 0, g(r(xs, j)) = \text{penalty}(xs, j)$                                                                                                         | (10.5): the penalty of the last job, zero for the empty schedule                                                                                                                                                                                                                                                                                       |
| 14 | $m$                         | <code>F[Job] → Int</code>                    | $m \triangleq [\text{zero}, \pi_1 \text{ cost}]$                                                                                                                                                                              | (10.6), B&dM's $h$ renamed as in $(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j))$<br>$\implies (\mu X : \frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{bagify}^\circ}{\exists} \text{ est}(R) : \text{the cost of the schedule before the last job}$ |

**[nil, snoc] cost = F((cost, bagify)) k**

(10.3b)

the cost of a schedule is got from the cost and the bag of the schedule before its last job, and that job

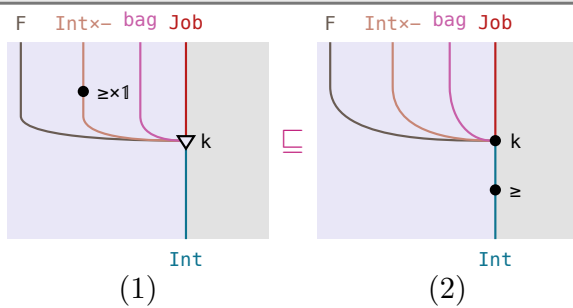


|   |                                    |                                                                                                                                                                                                                                                                          |
|---|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <code>[nil, snoc] cost</code>      | $\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$ , and $\text{penalty}(xs, j)$ reads only the bag of $xs$ — $\text{penalty}(xs, j) \triangleq (\text{sum}(\text{list}(\text{ct})) \text{ } xs + \text{ct } j - \text{dt } j) \text{wt}(j)$ |
| 2 | <code>= F((cost, bagify)) k</code> |                                                                                                                                                                                                                                                                          |

$$F(\geq \times 1) \ k \sqsubseteq k \geq$$

Exercise 10.5

a larger cost before the last job gives a larger cost after it

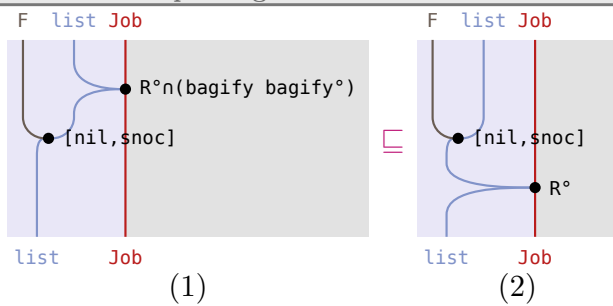


|   |                        |                                                 |
|---|------------------------|-------------------------------------------------|
| 1 | $F(\geq \times 1) \ k$ | $\text{bmax}$ is monotone in its first argument |
| 2 | $\sqsubseteq k \geq$   |                                                 |

$$F(R^\circ \ n \ (\text{bagify} \ \text{bagify}^\circ)) \ [\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}] \ R^\circ$$

(10.2)

improving the schedule before the last job, among schedules of the same bag, improves the whole schedule

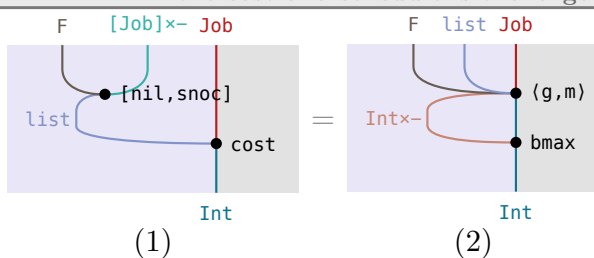


|   |                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $F(R^\circ \ n \ (\text{bagify} \ \text{bagify}^\circ)) \ [\text{nil}, \text{snoc}]$ | Proposition 9.3 at $S \triangleleft \text{bagify}$ , $\triangleleft \triangleleft \geq$ — Proposition 9.3<br>$\text{Map}(\text{cost}) \wedge \text{Simple}(S) \wedge h \ \text{cost} = F(\langle \text{cost}, S \rangle) \ k$<br>$\wedge F(\leq \times 1) \ k \sqsubseteq k \leq$<br>$\Rightarrow F(\langle \text{cost} \leq \text{cost}^\circ \rangle \ n \ (S \ S^\circ)) \ h \sqsubseteq h \ \text{cost} \leq$<br>$\text{cost}^\circ$ , with $[\text{nil}, \text{snoc}] \ \text{cost} = F(\langle \text{cost}, \text{bagify} \rangle) \ k$<br>and Exercise 10.5 $F(\geq \times 1) \ k \sqsubseteq k \geq$ |
| 2 | $\sqsubseteq [\text{nil}, \text{snoc}] \ R^\circ$                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

$$[\text{nil}, \text{snoc}] \ \text{cost} = \langle g, m \rangle \ \text{bmax}$$

(10.4)

the cost of a schedule is the larger of the penalty of its last job and the cost of the schedule before it

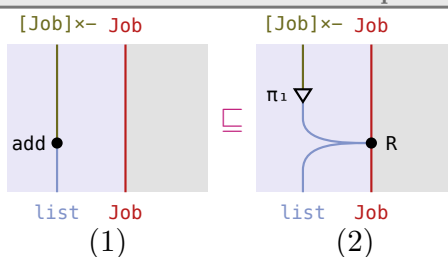


|   |                                           |                                                                                                                                                                                                                                                      |
|---|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $[\text{nil}, \text{snoc}] \ \text{cost}$ | $\text{cost} \ (xs \# [j]) = \text{bmax} \ (\text{cost} \ xs, \text{penalty} \ (xs, j))$ and<br>$\text{bmax}$ commutes — $\text{cost}(\text{wrap}(a)) = 0$ ,<br>$\text{cost}(\text{snoc}(x, a)) = \text{bmax}(\text{cost}(x), \text{penalty}(x, a))$ |
| 2 | $= \langle g, m \rangle \ \text{bmax}$    |                                                                                                                                                                                                                                                      |

$$(\forall j. \ 0 \leq \text{ct}(j)) \wedge (\forall j. \ 0 \leq \text{wt}(j)) \Rightarrow \text{add} \sqsubseteq \pi_1 \ R$$

(10.7)

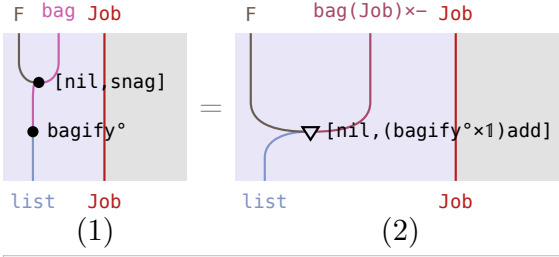
putting one more job anywhere into a schedule never lowers its cost



|   |                         |                                                                                      |
|---|-------------------------|--------------------------------------------------------------------------------------|
| 1 | $\text{add}$            | induction on where $j$ goes: every job after it starts later, so no<br>penalty falls |
| 2 | $\sqsubseteq \pi_1 \ R$ |                                                                                      |

$$[\text{nil}, \text{snag}] \text{ bagify}^\circ = [\text{nil}, (\text{bagify}^\circ \times 1) \text{ add}] \quad (10.8)$$

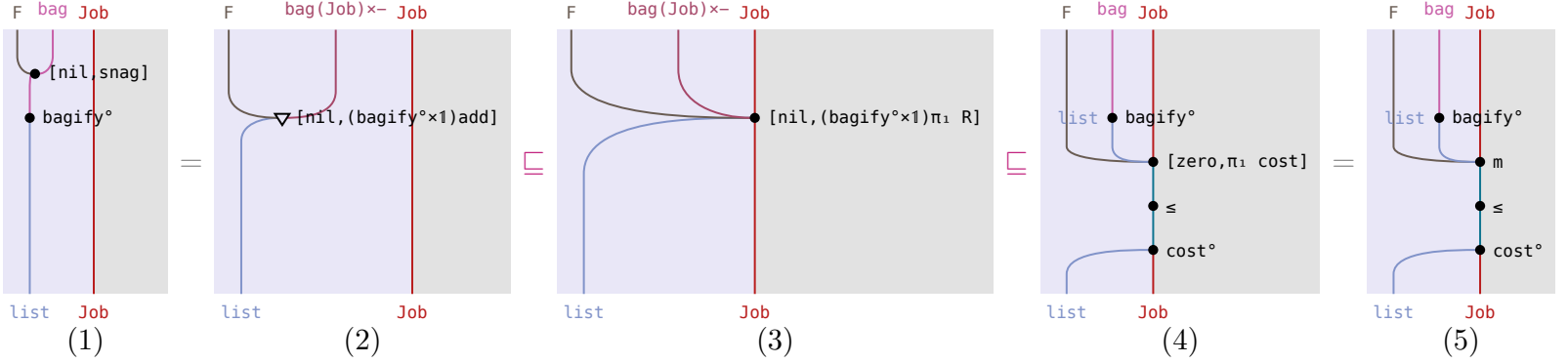
the orderings of a bag with one more job are the orderings of the bag with that job put in anywhere



|   |                                                              |                                                                          |
|---|--------------------------------------------------------------|--------------------------------------------------------------------------|
| 1 | $[\text{nil}, \text{snag}] \text{ bagify}^\circ$             | take out the last copy of $j$ and put it back — $p \text{ add } ys \iff$ |
| 2 | $= [\text{nil}, (\text{bagify}^\circ \times 1) \text{ add}]$ | $\text{add}(\pi_2(p), ys)$                                               |

$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \implies [\text{nil}, \text{snag}] \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) m \leq \text{cost}^\circ \quad (10.3c)$$

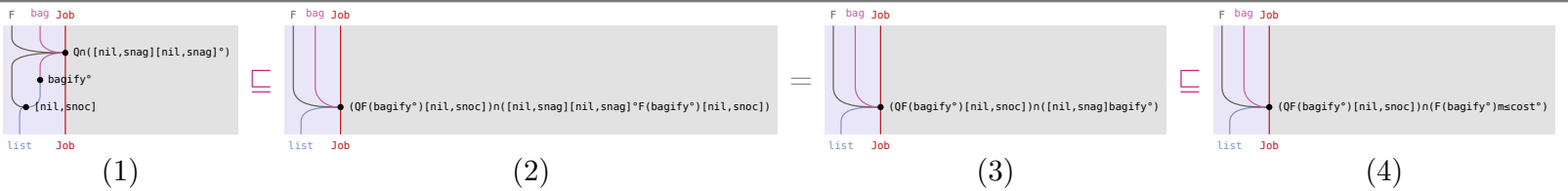
an ordering of a bag with one more job costs at least the cost of the ordering of the bag before its last job



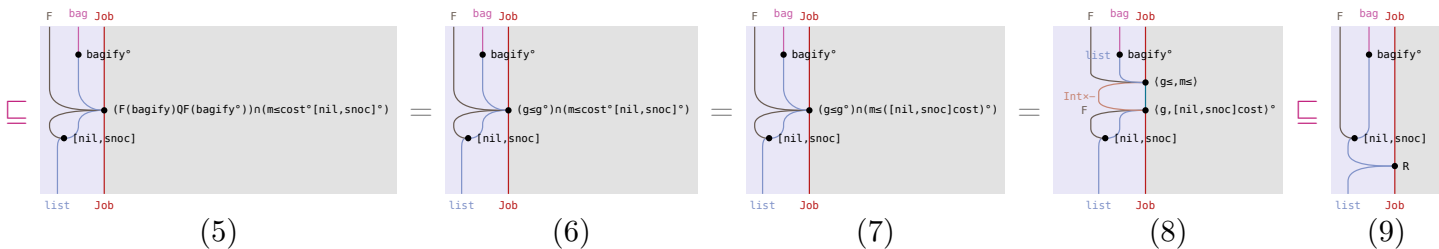
|   |                                                                                               |                                                                                                                           |
|---|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1 | $[\text{nil}, \text{snag}] \text{ bagify}^\circ$                                              | (10.8) $[\text{nil}, \text{snag}] \text{ bagify}^\circ = [\text{nil}, (\text{bagify}^\circ \times 1) \text{ add}]$        |
| 2 | $= [\text{nil}, (\text{bagify}^\circ \times 1) \text{ add}]$                                  | (10.7) $(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \implies \text{add} \sqsubseteq \pi_1 R$ |
| 3 | $\sqsubseteq [\text{nil}, (\text{bagify}^\circ \times 1) \pi_1 R]$                            | $[\text{nil}, \pi_1 R] \sqsubseteq [\text{zero}, \pi_1 \text{ cost}] \leq \text{cost}^\circ$                              |
| 4 | $\sqsubseteq F(\text{bagify}^\circ) [\text{zero}, \pi_1 \text{ cost}] \leq \text{cost}^\circ$ |                                                                                                                           |
| 5 | $= F(\text{bagify}^\circ) m \leq \text{cost}^\circ$                                           |                                                                                                                           |

$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \implies (Q n ([\text{nil}, \text{snag}] [\text{nil}, \text{snag}]^\circ)) F(\text{bagify}^\circ) [\text{nil}, \text{snoc}] \sqsubseteq F(\text{bagify}^\circ) [\text{nil}, \text{snoc}] R \quad (10.3)$$

a schedule of a bag ending in a job of least penalty is no worse than any schedule of the same bag



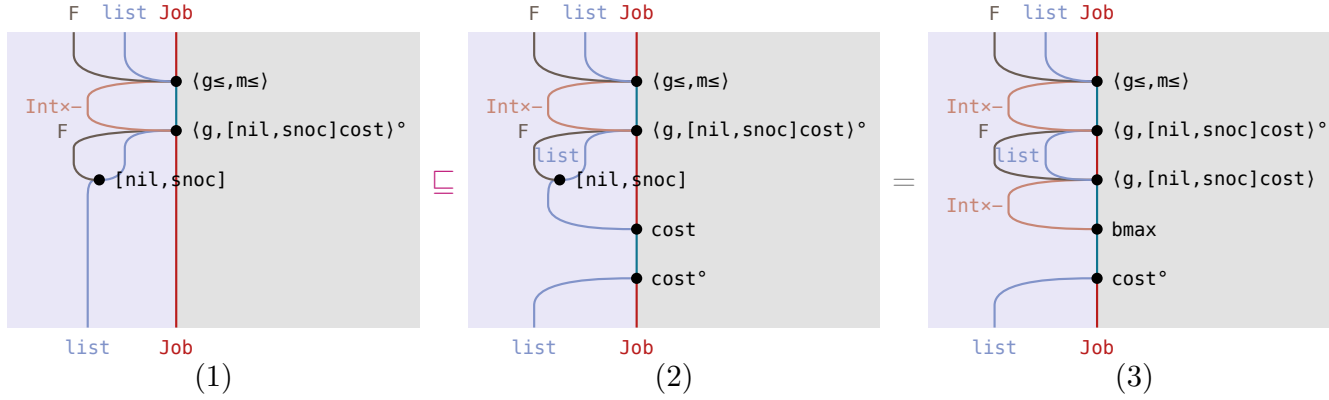
|   |                                                                                                                                                                                  |                                                                                                                                                                                                                                            |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $(Q n ([\text{nil}, \text{snag}] [\text{nil}, \text{snag}]^\circ)) F(\text{bagify}^\circ) [\text{nil}, \text{snoc}]$                                                             | $(S n T) R \sqsubseteq (S R) n (T R)$                                                                                                                                                                                                      |
| 2 | $\sqsubseteq (Q F(\text{bagify}^\circ) [\text{nil}, \text{snoc}] n ([\text{nil}, \text{snag}] [\text{nil}, \text{snag}]^\circ F(\text{bagify}^\circ) [\text{nil}, \text{snoc}])$ | $[\text{nil}, \text{snag}]^\circ F(\text{bagify}^\circ) [\text{nil}, \text{snoc}] = \text{bagify}^\circ$                                                                                                                                   |
| 3 | $= (Q F(\text{bagify}^\circ) [\text{nil}, \text{snoc}]) n ([\text{nil}, \text{snag}] \text{ bagify}^\circ)$                                                                      | (10.3c) $(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \implies [\text{nil}, \text{snag}] \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) m \leq \text{cost}^\circ$                                     |
| 4 | $\sqsubseteq (Q F(\text{bagify}^\circ) [\text{nil}, \text{snoc}]) n (F(\text{bagify}^\circ) m \leq \text{cost}^\circ)$                                                           | $(Q F(\text{bagify}^\circ) [\text{nil}, \text{snoc}]) n (F(\text{bagify}^\circ) M) \sqsubseteq F(\text{bagify}^\circ) ((F(\text{bagify}^\circ) Q F(\text{bagify}^\circ)) n (M [\text{nil}, \text{snoc}]^\circ)) [\text{nil}, \text{snoc}]$ |



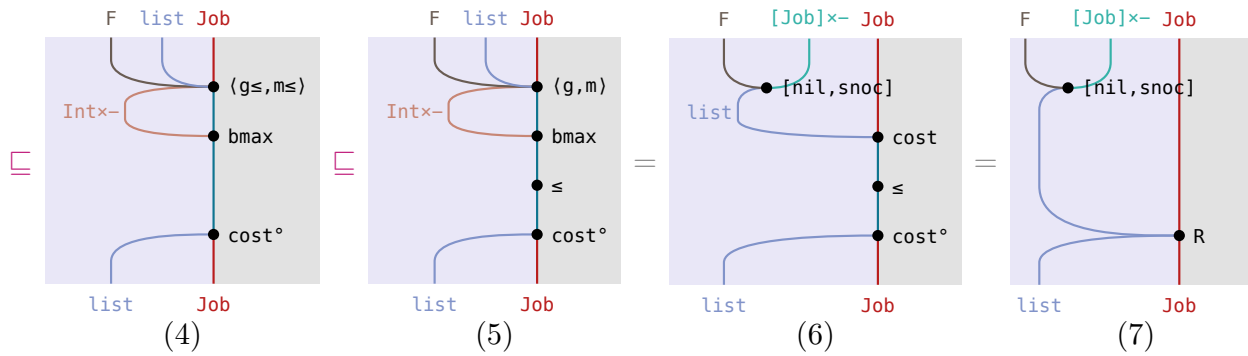
|   |                                                                                                                                                                                 |                                                                                                                |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 5 | $\sqsubseteq F(\text{bagify}^\circ) ((F(\text{bagify}^\circ) Q F(\text{bagify}^\circ)) n (m \leq \text{cost}^\circ [\text{nil}, \text{snoc}]^\circ)) [\text{nil}, \text{snoc}]$ | $F(\text{bagify}^\circ) Q F(\text{bagify}^\circ) = g \leq g^\circ$                                             |
| 6 | $= F(\text{bagify}^\circ) ((g \leq g^\circ) n (m \leq \text{cost}^\circ [\text{nil}, \text{snoc}]^\circ)) [\text{nil}, \text{snoc}]$                                            | $(R S)^\circ = S^\circ R^\circ$                                                                                |
| 7 | $= F(\text{bagify}^\circ) ((g \leq g^\circ) n (m \leq ([\text{nil}, \text{snoc}] \text{ cost}^\circ)^\circ)) [\text{nil}, \text{snoc}]$                                         | $\langle X, Y \rangle \langle R, S \rangle^\circ = \langle X R^\circ \rangle n \langle Y S^\circ \rangle$      |
| 8 | $= F(\text{bagify}^\circ) \langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost}^\circ \rangle^\circ [\text{nil}, \text{snoc}]$                       | (10.3d) $\langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost}^\circ \rangle^\circ$ |
| 9 | $\sqsubseteq F(\text{bagify}^\circ) [\text{nil}, \text{snoc}] R$                                                                                                                | $[\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}] R$                                            |

$$\langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ [\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}] R \quad (10.3d)$$

a job whose penalty is at most the last penalty, put after a schedule costing at most the cost before the last job,  
gives a schedule costing at most the whole



|   |                                                                                                                                                                                      |                                                                                                                                           |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ [\text{nil}, \text{snoc}]$                                                           | $1 \sqsubseteq \text{cost cost}^\circ$                                                                                                    |
| 2 | $\sqsubseteq \langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ [\text{nil}, \text{snoc}] \text{ cost cost}^\circ$                       | $[\text{nil}, \text{snoc}] \text{ cost} = \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle \text{ bmax}$                         |
| 3 | $= \langle g \leq, m \leq \rangle \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle \text{ bmax cost}^\circ$ | $\langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle^\circ \langle g, [\text{nil}, \text{snoc}] \text{ cost} \rangle \sqsubseteq 1$ |



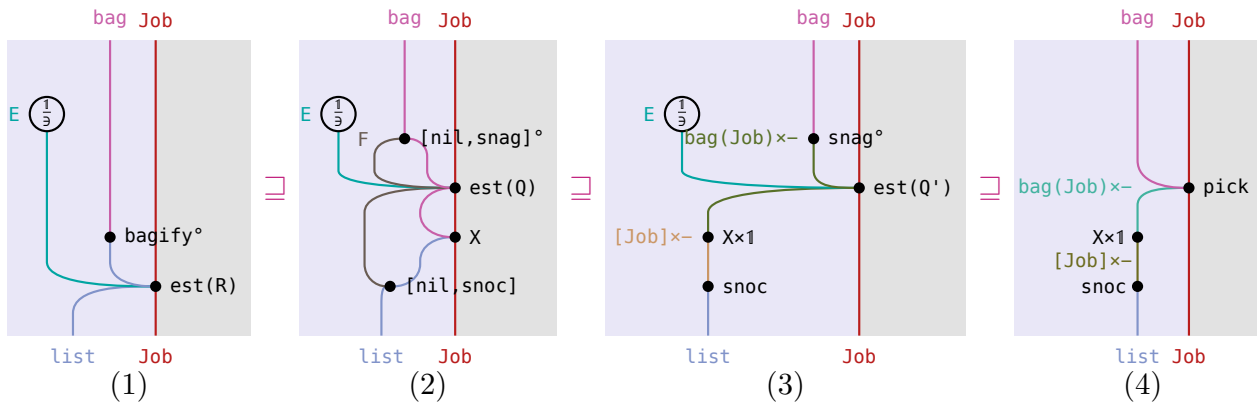
|   |                                                                        |                                                                                                  |
|---|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 4 | $\sqsubseteq \langle g \leq, m \leq \rangle \text{ bmax cost}^\circ$   | $\langle g \leq, m \leq \rangle \text{ bmax} \sqsubseteq \langle g, m \rangle \text{ bmax} \leq$ |
| 5 | $\sqsubseteq \langle g, m \rangle \text{ bmax} \leq \text{cost}^\circ$ | (10.4) $[\text{nil}, \text{snoc}] \text{ cost} = \langle g, m \rangle \text{ bmax}$              |
| 6 | $= [\text{nil}, \text{snoc}] \text{ cost} \leq \text{cost}^\circ$      | $R = \text{cost} \leq \text{cost}^\circ$                                                         |
| 7 | $= [\text{nil}, \text{snoc}] R$                                        |                                                                                                  |

$$\text{schedule} = (\text{null} \rightarrow \text{nil}, \text{pick} (\text{schedule} \times 1) \text{ snoc}), (\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (\mu X : \quad (10.3e)$$

$$\frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}] \sqsubseteq \frac{\text{bagify}^\circ}{\exists} \text{ est}(R)$$

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag,  
putting it last, and scheduling what is left the same way

the specification —  $\text{bagify} \triangleq \text{bagify}$



|   |                                                                                                             |                                                                                                                                                                    |
|---|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\frac{\text{bagify}^\circ}{\exists} \text{ est}(R)$                                                        | No greedy <b>reduce</b> exists — one would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole |
| 2 | $\sqsubseteq \frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}]$ | Proposition 10.1: <b>nil</b> and <b>snag</b> have disjoint ranges                                                                                                  |
| 3 | $\sqsubseteq \frac{\text{snag}^\circ}{\exists} \text{ est}(Q') (X \times 1) \text{ snoc}$                   | $\text{pick} \sqsubseteq \frac{\text{snag}^\circ}{\exists} \text{ est}(Q')$ , a partial function, quadratic in the number of jobs                                  |
| 4 | $\sqsubseteq \text{pick} (X \times 1) \text{ snoc}$                                                         |                                                                                                                                                                    |

## §10.4. The TeX problem

(10.4a)

|    | name         | type                                                              | definition                                                                                                                   | meaning                                                                           |
|----|--------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1  | intern       | $[\text{Digit}] \rightarrow [0, 2^{16})$                          | $\text{intern} \triangleq \text{val round}$                                                                                  | the value of a decimal fraction, rounded to 16 binary places                      |
| 2  | val          | $[\text{Digit}] \rightarrow \text{Real}$                          | $\text{val} \triangleq ([\text{zero}, \text{shift}])$                                                                        | the real number a digit list denotes after the decimal point                      |
| 3  | shift        | $\text{Digit} \times \text{Real} \rightarrow \text{Real}$         | $\text{shift} \triangleq \text{shift}, \text{shift}((d, r)) \triangleq (d + r)/10$                                           | put digit $d$ in front of $r$ : $(d+r)/10$                                        |
| 4  | round        | $\text{Real} \rightarrow [0, 2^{16})$                             | $r \text{ round } n \iff (2n-1)/w < r \wedge r < (2n+1)/w$ ,<br>$\text{round}^\circ = \text{interval inrange}$               | the nearest integer to $2^{16}r$ : round $r=n$ iff $2n-1 < 2^{17}r < 2n+1$        |
| 5  | interval     | $[0, 2^{16}) \rightarrow \text{Interval}$                         | $\text{interval}(n) \triangleq ((2n-1)/w, (2n+1)/w)$                                                                         | the open interval $((2n-1)/2^{17}, (2n+1)/2^{17})$ of the reals that round to $n$ |
| 6  | inrange      | $\text{Interval} \rightarrow \text{Real}$                         | $p \text{ inrange } r \iff \text{lo } p < r \wedge r < \text{hi } p$                                                         | $r$ lies strictly inside $(a, b)$                                                 |
| 7  | Legal( $p$ ) | Prop                                                              | $\text{Legal}(p) \triangleq 0 < \text{hi } p \wedge \text{hi } p < 1 \wedge \text{lo } p < \text{hi } p$                     | (10.9): $0 < b < 1$ and $a < b$ ; Interval is the legal pairs                     |
| 8  | arb          | $1 \rightarrow \text{Interval}$                                   | $x \text{ arb } p \iff \text{lo } p < 0 \wedge 0 < \text{hi } p$                                                             | any legal interval                                                                |
| 9  | step         | $\text{Digit} \times \text{Interval} \rightarrow \text{Interval}$ | $\text{step}(p) \triangleq (\text{shift}((\pi_1(p), \text{lo}(\pi_2(p)))) , \text{shift}((\pi_1(p), \text{hi}(\pi_2(p)))) )$ | put digit $d$ in front of both ends: $((d+a)/10, (d+b)/10)$                       |
| 10 | H            | $\text{Interval} \rightarrow [\text{Digit}]$                      | $H \triangleq ([\text{arb}, \text{step}])^\circ$                                                                             | every digit list whose interval the given one is                                  |
| 11 | R            | $[\text{Digit}] \rightarrow [\text{Digit}]$                       | $xs \text{ R } ys \iff \text{length}(xs) \leq \text{length}(ys)$                                                             | $x$ is no longer than $y$                                                         |
| 12 | l            | $1 \rightarrow \text{FX}$                                         | $l \triangleq \text{inl}$                                                                                                    | the injection of the empty case into $\text{FX} = 1 + (\text{Digit} \times X)$    |
| 13 | r            | $\text{Digit} \times X \rightarrow \text{FX}$                     | $r \triangleq \text{inr}$                                                                                                    | the injection of the digit case into $\text{FX} = 1 + (\text{Digit} \times X)$    |
| 14 | !            | $\text{Digit} \times \text{Interval} \rightarrow 1$               | $! \triangleq \rightarrow ()$                                                                                                | the one map to 1                                                                  |
| 15 | Q            | $F(\text{Interval}) \rightarrow F(\text{Interval})$               | $Q \triangleq (l^\circ !^\circ r) \cup 1, Q(u, z) \iff (\exists t. u = l(t) \wedge (\exists s. z = r(s))) \vee u = z$        | $u$ is the empty case and $z$ a digit case, or $u=z$                              |
| 16 | w            | Int                                                               | $w \triangleq 131072$                                                                                                        | $2^{17}$                                                                          |

 $([arb, step]) \sqsubseteq \text{val inrange}^\circ$ 

(10.4b)

every interval got by folding a decimal's digits with  $[arb, step]$  has the decimal's value strictly inside it

|   |                                                             |   |                                                                               |               |  |   |  |   |  |
|---|-------------------------------------------------------------|---|-------------------------------------------------------------------------------|---------------|--|---|--|---|--|
|   |                                                             | = |                                                                               | $\sqsubseteq$ |  | = |  | = |  |
| 1 | $F(\text{inrange}^\circ) [arb, step]$                       |   | $F(S) [T, U] = [T, (1 \times S) U]$                                           |               |  |   |  |   |  |
| 2 | $= [arb, (1 \times \text{inrange}^\circ) step]$             |   | $(1 \times \text{inrange}^\circ) step \sqsubseteq \text{shift inrange}^\circ$ |               |  |   |  |   |  |
| 3 | $\sqsubseteq [arb, \text{shift inrange}^\circ]$             |   | $\text{zero inrange}^\circ = \text{arb}$                                      |               |  |   |  |   |  |
| 4 | $= [\text{zero inrange}^\circ, \text{shift inrange}^\circ]$ |   | $[R, S] Z = [R Z, S Z]$                                                       |               |  |   |  |   |  |
| 5 | $= [\text{zero}, \text{shift}] \text{ inrange}^\circ$       |   |                                                                               |               |  |   |  |   |  |

$$Q \ F(X) \ \alpha \sqsubseteq F(X) \ \alpha \ R$$

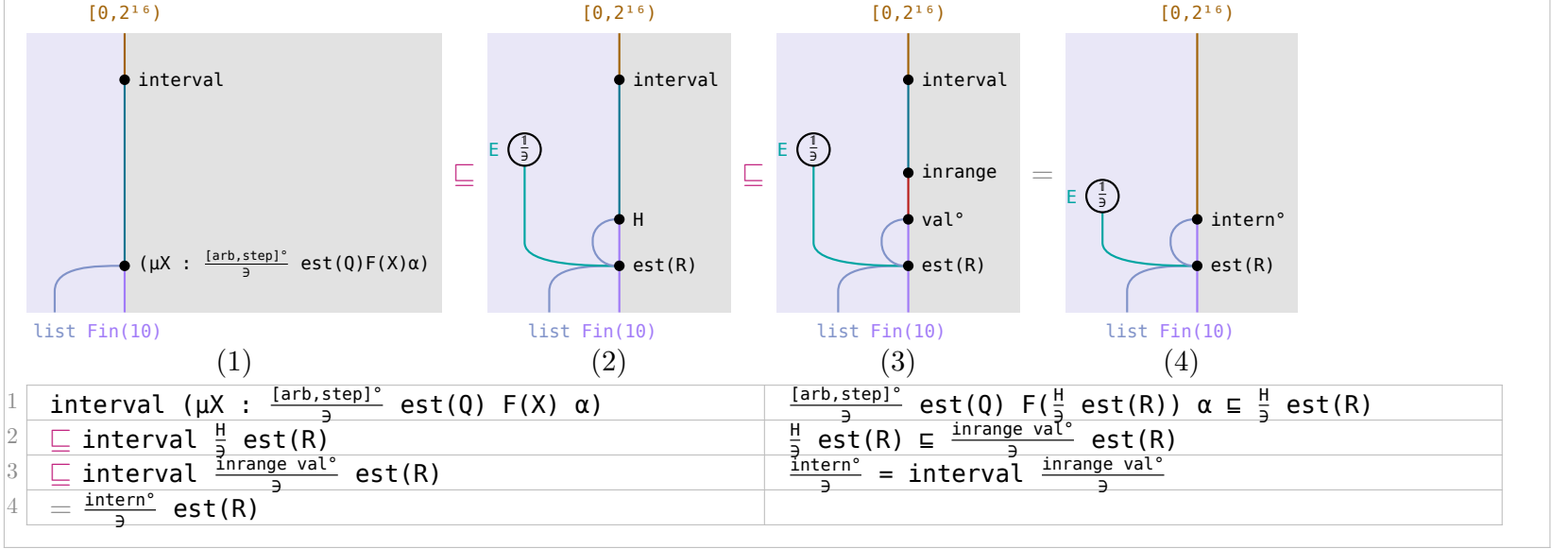
(10.4c)

choosing by  $Q$  among the one-step decompositions before building with  $F(X)$  and  $\alpha$  yields only decimals that are  $R$ -related to one built without choosing

|                                                                                                                                                                                                                                                                        |                                |                                                                                                                                   |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <p>list Fin(10)</p> <p>(1)</p>                                                                                                                                                                                                                                         | <p>list Fin(10)</p> <p>(2)</p> | <p>list Fin(10)</p> <p>(3)</p>                                                                                                    | <p>list Fin(10)</p> <p>(4)</p> |
| $\begin{array}{l} 1 \quad Q \ F(X) \ \alpha \\ 2 \quad \triangleq ((l^o \ !^o \ r) \ u \ 1) \ F(X) \ \alpha \\ 3 \quad = (l^o \ !^o \ r \ F(X) \ \alpha) \ u \ (F(X) \ \alpha) \end{array}$                                                                            |                                | $\begin{array}{l} Q \triangleq (l^o \ !^o \ r) \ u \ 1 \\ (S \ u \ T) \ R = (S \ R) \ u \ (T \ R) \\ 1 \sqsubseteq R \end{array}$ |                                |
| <p>list Fin(10)</p> <p>(5)</p>                                                                                                                                                                                                                                         | <p>list Fin(10)</p> <p>(6)</p> |                                                                                                                                   |                                |
| $\begin{array}{l} 4 \quad \sqsubseteq (l^o \ !^o \ r \ F(X) \ \alpha) \ u \ (F(X) \ \alpha \ R) \\ 5 \quad \sqsubseteq (l^o \ !^o \ ! \ l \ \alpha \ R) \ u \ (F(X) \ \alpha \ R) \\ 6 \quad \sqsubseteq (l^o \ l \ \alpha \ R) \ u \ (F(X) \ \alpha \ R) \end{array}$ |                                | $\begin{array}{l} r \ F(X) \ \alpha \sqsubseteq ! \ l \ \alpha \ R \\ !^o \ ! \sqsubseteq 1 \\ l \ F(X) = l \end{array}$          |                                |
| <p>list Fin(10)</p> <p>(7)</p>                                                                                                                                                                                                                                         | <p>list Fin(10)</p> <p>(8)</p> | <p>list Fin(10)</p> <p>(9)</p>                                                                                                    |                                |
| $\begin{array}{l} 7 \quad = (l^o \ l \ F(X) \ \alpha \ R) \ u \ (F(X) \ \alpha \ R) \\ 8 \quad \sqsubseteq (F(X) \ \alpha \ R) \ u \ (F(X) \ \alpha \ R) \\ 9 \quad = F(X) \ \alpha \ R \end{array}$                                                                   |                                | $\begin{array}{l} l^o \ l \sqsubseteq 1 \\ R \ u \ R = R \end{array}$                                                             |                                |

$$\text{extern}(n)=f(2n-1,2n+1), \quad \text{interval } (\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ est}(Q) F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{ est}(R) \quad (10.4d)$$

a shortest decimal whose internal representation is the given multiple of  $2^{-16}$  is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do



- $f(a,b) = xs \iff a < 0 \wedge xs = [] \vee \neg a < 0 \wedge (\exists d. \exists ys. d = [10b] \wedge f(10a-d, 10b-d) = ys \wedge xs = [d] \# ys)$   
 $f$  gives  $(a,b)$  the empty decimal when  $a < 0$ , and otherwise the digit  $d = [10b]$  followed by what it gives  $(10a-d, 10b-d)$  (10.4e)
- $(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ est}(Q) F(X) \alpha) = f$   
the least solution of the greedy recursion of  $\text{interval } (\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ est}(Q) F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{ est}(R)$  is  $f$
- $d = [10b] \wedge e = [10b] \implies d = e$   
Exercise 10.12: two digits  $d$  and  $e$  with  $0 < 10b-d < 1$  and  $0 < 10b-e < 1$  are equal
- $(0 < q-p \wedge q < w) \wedge p < 0 \implies f(p,q) = []$   
the program's first clause:  $f(p,q) = []$  when  $p < 0$
- $(0 < q-p \wedge q < w) \wedge \neg p < 0 \implies f(p,q) = [10q/w] \# f(10p-w \cdot (10q/w), 10q-w \cdot (10q/w))$   
the program's second clause: with  $d = (10q) \text{ div } w$ ,  $f(p,q)$  is  $d$  followed by  $f(10p-w \cdot d, 10q-w \cdot d)$
- $(0 < q-p \wedge q < w) \wedge \text{Reach}(p,q) \implies f(p/w, q/w) = f(p,q)$   
on the pairs the program reaches from  $\text{interval } n$ , the integer  $f$  at  $(p,q)$  is a value of the rational  $f$  at  $(p/w, q/w)$
- $\text{interval } f = \text{extern}$   
 $f$  after  $\text{interval}$  is the program  $\text{extern}$ , whose value at  $n$  is  $f(2n-1, 2n+1)$

## §10. Appendix

### §10.1. $P(S) \text{ est}(R) = (\exists S) \cap (\epsilon \setminus (SR^\circ))$

(10.1a)

$(p, q)$  tabulates  $W \triangleq (\exists S) \cap (\epsilon \setminus (SR^\circ))$ , and  $y \triangleq \frac{(p \exists S) \cap (qR)}{\exists}$ , a map.

(10.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} W \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = W, 1 \sqsubseteq yy^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(10.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f^\circ, \cdot \exists \neg \exists} \boxed{\begin{array}{c} q \\ (p \exists S) \cap (qR) \end{array}} \xleftarrow{\Delta \neg \cap, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

$$\boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \exists, ^\circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple}$$

(10.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg \cap, \cdot T \neg / T, ^\circ, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array}$$

$$\boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \exists, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple}$$

$$\boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) \cap (qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq qq^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{^\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} W \\ \epsilon \setminus (SR^\circ) \end{array}} \quad \text{W's right half}$$

**§10.2.  $P(\text{est}(R)) \text{ est}(R) = P(\text{dom}(\text{est}(R))) \text{ union est}(R)$** 

(10.2a)

$$\boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)}$$

$$\begin{array}{ccc} \xrightarrow{\text{P a relator}} \boxed{P(\text{dom}(\text{est}(R)))P(\text{est}(R)) \text{ est}(R)} & \xrightarrow{(7.11) \text{ } R \circ R \subseteq R} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \\ \Rightarrow P(\text{est}(R)) \text{ est}(R) \subseteq \text{union est}(R) & & \end{array}$$

(10.2b)

$$\begin{array}{ccc} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} & \xleftarrow{1 \subseteq R} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} & \xleftarrow{\exists \text{est}(R)} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \\ \boxed{P(\text{est}(R)) \text{ est}(R)} & \Rightarrow P(S) \text{ est}(R) = (\exists S) \cap (\epsilon \setminus (S \circ R^\circ)) \text{ at } S := \text{est}(R), & \boxed{\epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \\ & \Delta \vdash n, T \vdash \neg T \setminus & \boxed{\text{est}(R) R^\circ} \end{array}$$

$$\boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \xleftarrow{P(\text{dom}(\text{est}(R))) \subseteq 1} \boxed{\text{union est}(R)}$$

$$\begin{array}{ccc} \xleftarrow{T(\text{UnV}) \subseteq T \text{UnTV}, \cdot \exists \vdash \frac{}{\exists}, (7.3) \text{ union } (\epsilon \setminus R) = \epsilon \setminus (\epsilon \setminus R)} \boxed{(\exists \exists) n(\epsilon \setminus (\epsilon \setminus R^\circ))} & \xleftarrow{\text{modular law}} \boxed{\exists(\exists n(\epsilon \setminus (\epsilon \setminus R^\circ)))} \\ \boxed{\exists \text{est}(R)} & & \boxed{\exists(\exists n(\epsilon \setminus R^\circ))} \\ & & \text{count of } T \cdot \neg T \setminus \end{array}$$

$$\boxed{\epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\text{dom}(\text{est}(R)) \epsilon \text{ union est}(R)}$$

$$\begin{array}{ccc} \xleftarrow{\text{dom}(\text{est}(R)) \subseteq \text{est}(R) \text{ est}(R)^\circ} \boxed{\text{est}(R) \text{ est}(R)^\circ \epsilon \text{ union est}(R)} & \xleftarrow{\text{est}(R)^\circ \subseteq \epsilon, \epsilon \epsilon \text{ union} \subseteq \epsilon} \boxed{\text{est}(R) \epsilon \text{ est}(R)} \\ \boxed{\text{est}(R) R^\circ} & & \boxed{\text{est}(R) R^\circ} \\ & & \text{UP of est} \end{array}$$

## §10.3. Shortest paths on a cylinder, on lists

(10.3a)

|    | definition                                                                                                          | type                                       | note                                                                                                                                                                                                                |
|----|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | $F(A,X)=A+A \times X$                                                                                               | $\mathcal{A} \rightarrow \mathcal{A}$      | The base functor: one square of the new column, alone or in front of the path so far.                                                                                                                               |
| 2  | $L = \text{list}^+$                                                                                                 | $\mathcal{A} \rightarrow \mathcal{A}$      | A path is a non-empty list of squares, one per column crossed.                                                                                                                                                      |
| 3  | $\alpha$ the initial algebra                                                                                        | $F(\mu F) \rightarrow \mu F$               | $\alpha(5)=[5]$ and $\alpha(1,[5])=[1,5]$ : a path is started by one square, or extended by one.                                                                                                                    |
| 4  | $N$<br>the n-tuple relator                                                                                          | $\mathcal{A} \rightarrow \mathcal{A}$      | One component per row of a column, so the fold carries n answers at once.                                                                                                                                           |
| 5  | $R \triangleq \text{sum} \leq \text{sum}^\circ$                                                                     | $[\mathbb{N}] \rightarrow [\mathbb{N}]$    | The cost of a path, which the cheapest minimises.                                                                                                                                                                   |
| 6  | setify                                                                                                              | $A[n] \rightarrow PA$                      | $\text{setify}(1,2,3,4)=\{1,2,3,4\}$ — which row a component came from is forgotten.                                                                                                                                |
| 7  | moves                                                                                                               | $A[n] \rightarrow P(A[n])$                 | $\text{moves}(x)=\{\text{up}(x), x, \text{down}(x)\}$ — rotated up, unrotated, rotated down.                                                                                                                        |
| 8  | trans                                                                                                               | $P(A[n]) \rightarrow (P\ A)[n]$            | $\text{trans}\{(a,b,c), (x,y,z)\} = (\{a,x\}, \{b,y\}, \{c,z\})$ — component k of the result is the set of the k-th components.                                                                                     |
| 9  | zip                                                                                                                 | $F(a[n]) \rightarrow (F\ a)[n]$            | $\text{zip}((1,2,3,4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = ((1, \{[5]\}), (2, \{[6]\}), (3, \{[7]\}), (4, \{[8]\}))$ .                                                                                          |
| 10 | $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$                                                                | $F(PA) \rightarrow PFA$                    | $\text{cp}(1, \{[5], [6], [8]\}) = \{(1, [5]), (1, [6]), (1, [8])\}$ .                                                                                                                                              |
| 11 | $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union}))$<br>$\text{zip } N(\text{cp}(F(A, -))\ E(\alpha))$ | $F(NA, N(P(TA))) \rightarrow N(P(TA))$     | $\text{gen}((1,2,3,4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}))$ is worked out in $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -))\ E(\alpha))$ .                    |
| 12 | $\text{paths} \triangleq (\text{gen})\ \text{setify}\ \text{union}$                                                 | $T(NA) \rightarrow P(TA)$                  | $\text{paths}[(1,2,3,4), (5,6,7,8)]$ is the union of $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -))\ E(\alpha))$ 's four sets: 12 paths, 3 from each entry row. |
| 13 | the specification<br>$\text{paths est}(R)$                                                                          | $[\mathbb{N}[n]] \rightarrow [\mathbb{N}]$ | A cheapest path from the entry side to the exit side.                                                                                                                                                               |

§10.3.1.  $N(E(L\ A))$ 

(10.3.1a)

```

1  5
2  6      [(1,2,3,4), (5,6,7,8)] : L N Nat
3  7
4  8

```

(10.3.1b)

```

gen  $\triangleq$  F(1, moves trans N(union)) zip N(cp(F(A, -)) E( $\alpha$ ))
(gen)[(5,6,7,8)] = ({[5]}, {[6]}, {[7]}, {[8]})

gen((1,2,3,4), ({[5]}, {[6]}, {[7]}, {[8]})) =
( {[1,5]}, {[1,6]}, {[1,8]} ,
  {[2,5]}, {[2,6]}, {[2,7]} ,
  {[3,6]}, {[3,7]}, {[3,8]} ,
  {[4,5]}, {[4,7]}, {[4,8]} )

```

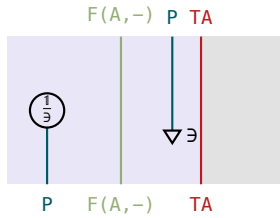
(10.3.1c)

|   |   |                                                        |
|---|---|--------------------------------------------------------|
| 1 | N | the 4-tuple, one component per row of the entry column |
| 2 | E | the set of paths that can start in that row            |
| 3 | L | a path is the non-empty list of the squares it crosses |

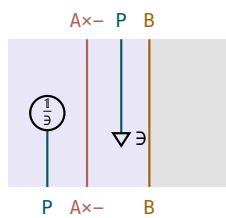
row 1's set has no  $[1,7]$  — from row 1 only rows 4, 1 and 2 are reachable — and it does have  $[1,8]$ , because the cylinder glues the bottom row to the top. That is  $\text{moves trans } N(\text{union})$  of  $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$  : component  $k$  collects the paths of rows  $k-1, k, k+1$ .

### §10.3.2. The cross product $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$

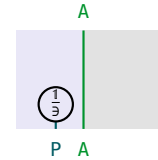
(10.3.2a)



cp on all of  $F(A, E(L A))$



the  $A \times -$  summand:  $\exists$  picks one path,  $1\% \exists$  collects the results



the  $A$  summand, no  $E$  to distribute:  $1\% \exists$  alone,  $a \mapsto \{a\}$

(10.3.2b)

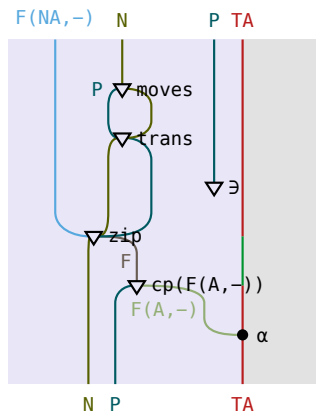
| type                                                                | note                                                                                                                        |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1 $F(A, -) = A + A \times -$                                        | one wire carries the whole functor, both summands with it; the $\exists$ under it is $F(1, \exists)$                        |
| 2 $\exists : E(L A) \rightarrow L A$                                | the $\exists$ inside $F(1, \exists)$ : a set of paths, one of them — $L A$ is one path, $A$ one square                      |
| 3 $F(1, \exists) = 1 + 1 \times \exists$                            | $: A + A \times E(L A) \rightarrow A + A \times L A$ , the relator acting on each summand                                   |
| 4 $\exists : E(A + A \times L A) \rightarrow A + A \times L A$      | the $\exists$ under the bar: a set of cells, one of them — a different $\exists$                                            |
| 5 $\text{cp} : A + A \times E(L A) \rightarrow E(A + A \times L A)$ | $\frac{R}{\exists}$ turns $R : X \rightarrow Y$ into $X \rightarrow E Y$                                                    |
| 6 $\text{cp} = \frac{1}{\exists} E(F(1, \exists))$                  | what the picture draws: the unit makes the outer $E$ , leaving one $\exists$ — $\frac{R}{\exists} = \frac{1}{\exists} E(R)$ |

(10.3.2c)

$w = (1, \{[5], [6], [8]\}) : A \times E(L A)$  the right summand  
 $1 \times \exists : A \times E(L A) \rightarrow A \times L A$   $1$  keeps the square 1,  $\exists$  picks one path  
 $w \mapsto (1, [5]), (1, [6]), (1, [8]) : A \times L A$  one output per path in  $w$   
 $\text{cp}(w) = \{(1, [5]), (1, [6]), (1, [8])\} : E(A + A \times L A)$  the three of them, collected

### §10.3.3. $\text{gen} = F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp } P(\alpha))$

(10.3.3a)

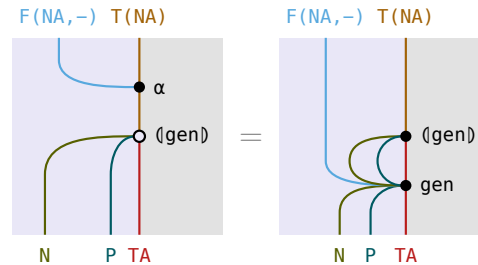


(10.3.3b)

$\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$   
 $u = ((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) : F(N A, N(E(L A)))$   
 $F(1, \text{moves trans } N(\text{union}))$  1 keeps the column, the path SETS move  
 $\text{moves}(\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$   
 $= (\{[6]\}, \{[7]\}, \{[8]\}, \{[5]\}),$   
 $(\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}),$   
 $(\{[8]\}, \{[5]\}, \{[6]\}, \{[7]\})$  :  $E(N(E(L A)))$  down, unmoved, up  
 $\text{trans}(\text{that})$   
 $= (\{[6]\}, \{[5]\}, \{[8]\}, \{[7]\}, \{[6]\}, \{[5]\}),$   
 $\{[8]\}, \{[7]\}, \{[6]\}, \{[5]\}, \{[8]\}, \{[7]\})$  :  $N(E(E(L A)))$  row k gets rows k-1, k, k+1  
 $N(\text{union})(\text{that})$   
 $= (\{[5], [6], [8]\}, \{[5], [6], [7]\},$   
 $\{[6], [7], [8]\}, \{[5], [7], [8]\})$  :  $N(E(L A))$  every path into row k, none dropped  
 $\text{zip}(\text{that})$   
 $= ((1, \{[5], [6], [8]\}), (2, \{[5], [6], [7]\}),$   
 $(3, \{[6], [7], [8]\}), (4, \{[5], [7], [8]\}))$  :  $N(F(A, E(L A)))$  each row: its square, and the paths it may be put in front of  
 $N(\text{cp } P(\alpha))(\text{that})$   
 $= (\{[1, 5], [1, 6], [1, 8]\}, \{[2, 5], [2, 6], [2, 7]\},$   
 $\{[3, 6], [3, 7], [3, 8]\}, \{[4, 5], [4, 7], [4, 8]\})$  :  $N(E(L A))$  cp pairs the square with each path,  $\alpha$  prefixes it:  $\alpha(1, [5]) = [1, 5]$

#### §10.3.4. $\text{gen}$ is an F-algebra; $(\text{gen}) : \alpha(\text{gen}) = F(1, (\text{gen})) \text{gen}$

(10.3.4a)

 $\alpha$  puts the column back on the list, then the fold reads all of itthe fold reads the rest under  $F$ , then one  $\text{gen}$  puts the column in front

$$xs = [(1, 2, 3, 4), (5, 6, 7, 8)] = \alpha((1, 2, 3, 4), [(5, 6, 7, 8)]) : L(N A)$$

(10.3.4b)

$$(\text{gen})(xs) = \text{gen}((1, 2, 3, 4), (\text{gen})[(5, 6, 7, 8)])$$

both halves are in  $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$  : the tail folds to  $(\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$ , one path per row and each of them one square long, and  $\text{gen}$  on it is  $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$  's walk.

#### §10.3.5. $Q : F(N A, N(L A)) \rightarrow N(L A)$ , an $F(N A, -)$ -algebra: $Q = F(1, \text{moves trans } N(\text{est}(R))) \text{ zip } N(\alpha)$

(10.3.5a)

$$Q : F(N A, N(L A)) \rightarrow N(L A)$$

the fold's algebra: a new column, one path per row

$u = ((1, 2, 3, 4), ([5], [6], [7], [8])) : F(N A, N(L A))$   
 $F(1, \text{moves trans } N(\text{est}(R)))$  1 keeps the column, the paths move  
 $\text{moves}([5], [6], [7], [8])$   
 $= ([6], [7], [8], [5]),$   
 $([5], [6], [7], [8]),$   
 $([8], [5], [6], [7])$  :  $E(N(L A))$  down, unmoved, up  
 $\text{trans}(\text{that})$   
 $= ([6], [5], [8]), ([7], [6], [5]),$   
 $[8], [7], [6]), ([5], [8], [7])$  :  $N(E(L A))$  row k gets rows k-1, k, k+1  
 $N(\text{est}(R))(\text{that})$   
 $= ([5], [5], [6], [5])$  :  $N(L A)$  the cheapest into each row – chosen BEFORE the new square  
 $\text{zip}(\text{that})$   
 $= ((1, [5]), (2, [5]), (3, [6]), (4, [5]))$  :  $N(F(A, L A))$  each row: its square, and the one predecessor that survived  
 $N(\alpha)(\text{that})$   
 $= ([1, 5], [2, 5], [3, 6], [4, 5])$  :  $N(L A)$   $\alpha(1, [5]) = [1, 5]$

$$F(1, \text{moves trans } N(\text{est}(R))) = [l, pm \ r] \wedge zip = [N(inl), zip' \ N(inr)] \wedge inl \ \alpha = wrap \wedge inr \ \alpha \quad (10.3.5b)$$

$$= \text{cons} \Rightarrow Q = [N(wrap), pm \ zip' \ N(\text{cons})]$$

at the last column  $Q$  starts one path per row, and at each earlier one it puts each square in front of the cheapest of the three kept paths it can step to — the algebra  $R \ R \sqsubseteq R \wedge F(1, N(\text{est}(R))) \ Q \sqsubseteq \text{gen } N(\text{est}(R))$

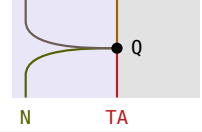
$\Rightarrow (Q)$  setify  $\text{est}(R) \sqsubseteq \text{paths } \text{est}(R)$  's last step folds

**circuit** — the fork is  $F(NA, N(LA)) = NA + NA \times N(LA)$

**Hinze–Marsden**

$$F(N \ A) \ (N \ (T \ A)) \rightarrow \boxed{Q} \rightarrow N(T \ A)$$

$$F(N \ A, N \ (T \ A))$$

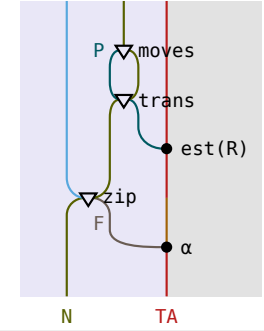


$$FN(T \ A) \rightarrow \boxed{F(1, \text{moves trans } N(\text{est}(R)))} \rightarrow FN(T \ A) \rightarrow \boxed{zip} \rightarrow N(F \ A \ (T \ A)) \rightarrow \boxed{N(\alpha)} \rightarrow N(T \ A)$$

= the fusion condition  $R \ R \sqsubseteq R \wedge F(1, \text{est}(R)) \ \alpha \sqsubseteq \text{cp}(F(A, -)) \ E(\alpha) \ \text{est}(R)$

$\Rightarrow F(1, N(\text{est}(R))) \ Q \sqsubseteq \text{gen } N(\text{est}(R))$  read as a definition,

$$F(NA, -) \ N \ TA$$



$$F(N \ A) \ (N \ (T \ A)) \rightarrow \left\{ \begin{array}{l} \boxed{l} \rightarrow N(p_0) \rightarrow \boxed{N(wrap)} \\ \boxed{r} \rightarrow q \rightarrow \boxed{pm} \rightarrow q \rightarrow \boxed{zip'} \rightarrow N(q_0) \rightarrow \boxed{N(cons)} \end{array} \right\} \rightarrow N(T \ A)$$

$$zip = 1 + zip', \ \alpha = [wrap, cons]$$

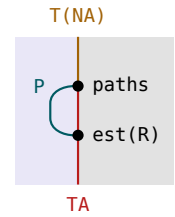
$$R \subseteq R \wedge F(1, N(\text{est}(R))) \quad Q \subseteq \text{gen } N(\text{est}(R)) \Rightarrow \langle Q \rangle \text{ setify } \text{est}(R) \subseteq \text{paths } \text{est}(R) \quad (10.3.5c)$$

shortest paths on a cylinder: the dynamic programming approach to exhaustive search allows a path of least cost to be found in  $O(n \times m)$  time;  $Q$  is  $F(1, \text{moves trans } N(\text{est}(R))) = [l, pm \ r] \wedge \text{zip} = [N(\text{inl}), \text{zip}' \ N(\text{inr})] \wedge \text{inl } \alpha = \text{wrap} \wedge \text{inr } \alpha = \text{cons}$   
 $\Rightarrow Q = [N(\text{wrap}), pm \ \text{zip}' \ N(\text{cons})]$  's algebra.

circuit

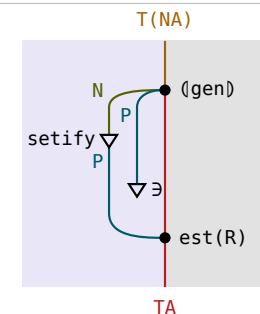
Hinze–Marsden

$$T(N \ A) \text{---} \boxed{\text{paths}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$



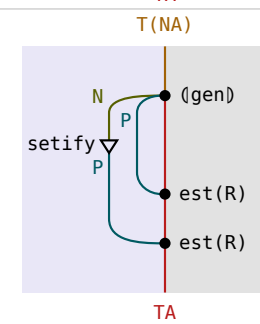
$$= T(N \ A) \text{---} \boxed{[F(N \ A) \ (N \ (P \ (T \ A)))]} \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{\text{setify}} \text{---} PP(T \ A) \text{---} \boxed{\text{union}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

$\text{paths} \triangleq \langle \text{gen} \rangle \text{ setify union at paths}$



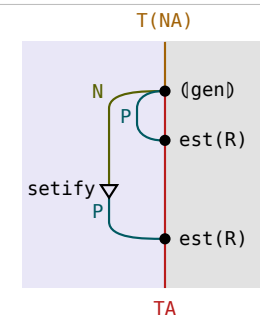
$$\sqsubseteq T(N \ A) \text{---} \boxed{[F(N \ A) \ (N \ (P \ (T \ A)))]} \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{\text{setify}} \text{---} PP(T \ A) \text{---} \boxed{P(\text{est}(R))} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

$P(\text{est}(R)) \text{ est}(R) \sqsubseteq \text{union est}(R)$  — a minimum in each set, then a minimum of those;  $R$  transitive



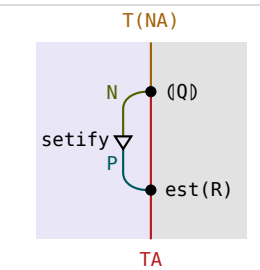
$$\sqsubseteq T(N \ A) \text{---} \boxed{[F(N \ A) \ (N \ (P \ (T \ A)))]} \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T \ A) \text{---} \boxed{\text{setify}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

$\text{setify lax natural}$



$$\sqsubseteq T(N \ A) \text{---} \boxed{[F(N \ A) \ (N \ (T \ A))]} \text{---} \boxed{Q} \text{---} N(T \ A) \text{---} \boxed{\text{setify}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

fusion at  $R \subseteq R \wedge F(1, \text{est}(R)) \quad \alpha \sqsubseteq \text{cp}(F(A, -)) \ E(\alpha) \ \text{est}(R)$   
 $\Rightarrow F(1, N(\text{est}(R))) \quad Q \subseteq \text{gen } N(\text{est}(R))$



| $R \subseteq R \wedge F(1, \text{est}(R)) \alpha \subseteq \text{cp}(F(A, -)) E(\alpha) \text{ est}(R) \Rightarrow F(1, N(\text{est}(R))) Q \subseteq \text{gen } N(\text{est}(R))$ (10.3.5d)<br>fusion: the condition for fusion in $R \subseteq R \wedge F(1, N(\text{est}(R))) Q \subseteq \text{gen } N(\text{est}(R))$<br>$\Rightarrow (Q) \text{ setify } \text{est}(R) \subseteq \text{paths } \text{est}(R)$ 's last step, used to derive a definition of $Q$ . |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Hinze–Marsden |
| $F(N(A) (N(P(T(A)))) \text{---} \boxed{\text{gen}} \text{---} NP(T(A)) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T(A))$                                                                                                                                                                                                                                                                                                                                          |               |
| $\sqsubseteq FNP(T(A)) \text{---} \boxed{F(1, N(\text{est}(R)))} \text{---} FN(T(A)) \text{---} \boxed{Q} \text{---} N(T(A))$ <p>(7.13), then zip, trans, moves lax natural</p>                                                                                                                                                                                                                                                                                         |               |

(7.13) is  $F(1, \text{est}(R)) \alpha \subseteq \text{cp } P(\alpha) \text{ est}(R)$ , Theorem 7.1  $\text{Map}(f) \wedge 1 \subseteq R$   
 $\Rightarrow f^\circ F(R) f \subseteq R \iff F(\text{est}(R)) f \subseteq \frac{F(\exists) f}{\exists} \text{ est}(R)$  at the map  $\alpha$  with  $\frac{F(1, \exists) \alpha}{\exists} = \text{cp } P(\alpha)$ : extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.